

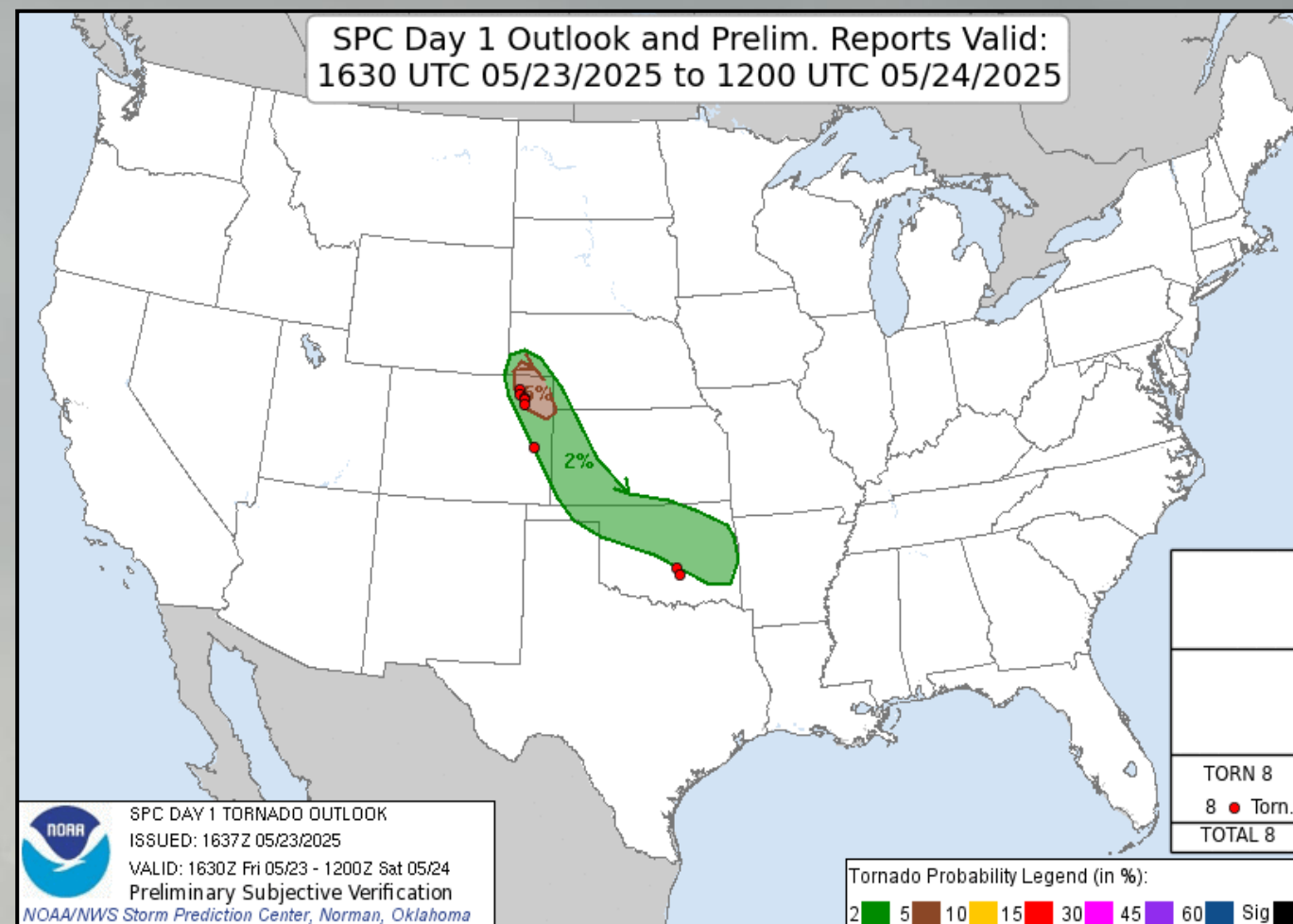
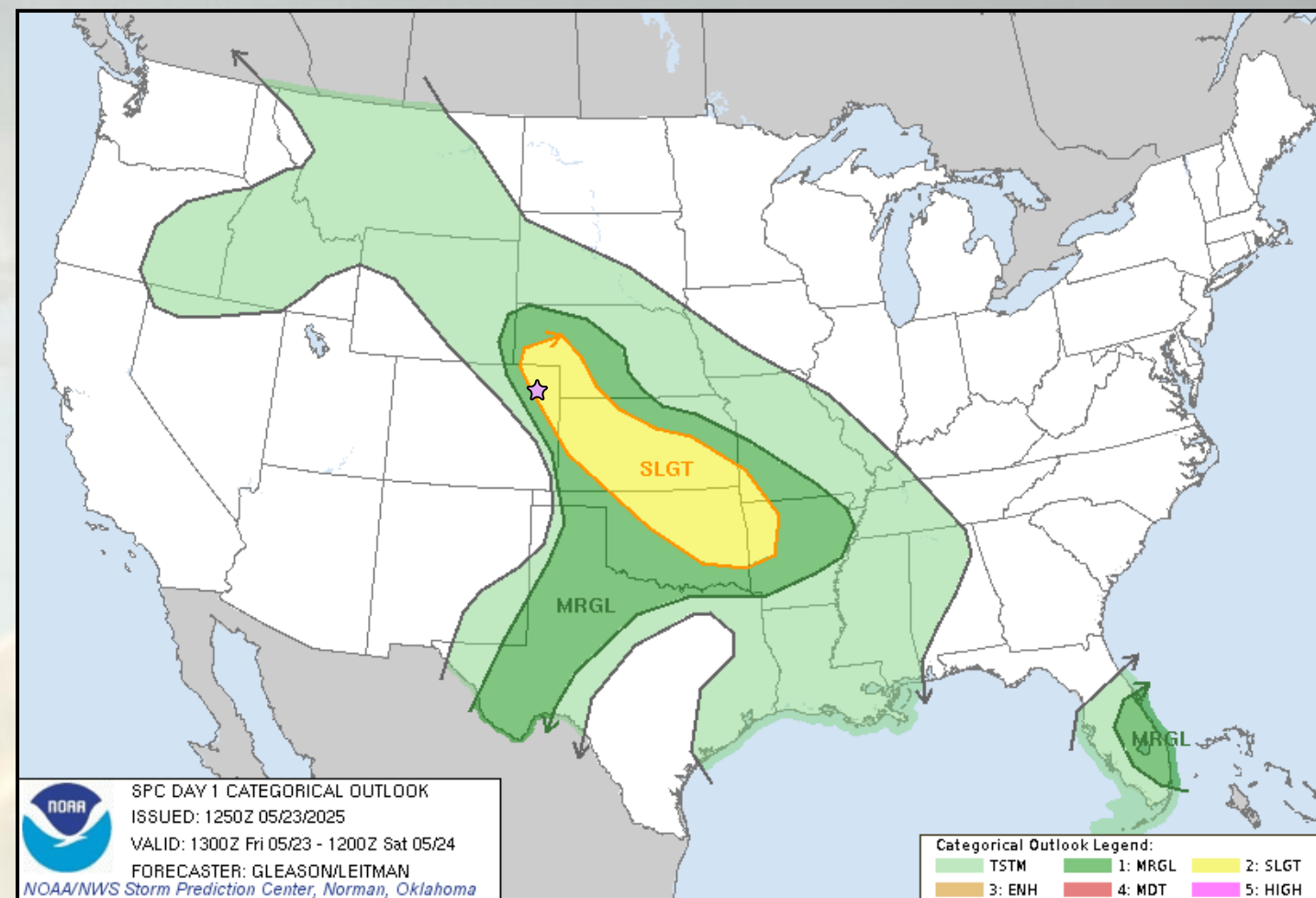
A SATELLITE & RADAR ANALYSIS OF THE MAY 23. 2025 TORNADO IN AKRON. COLORADO

(STORM CHASE)

Alexis Fugatt



EVENT OVERVIEW



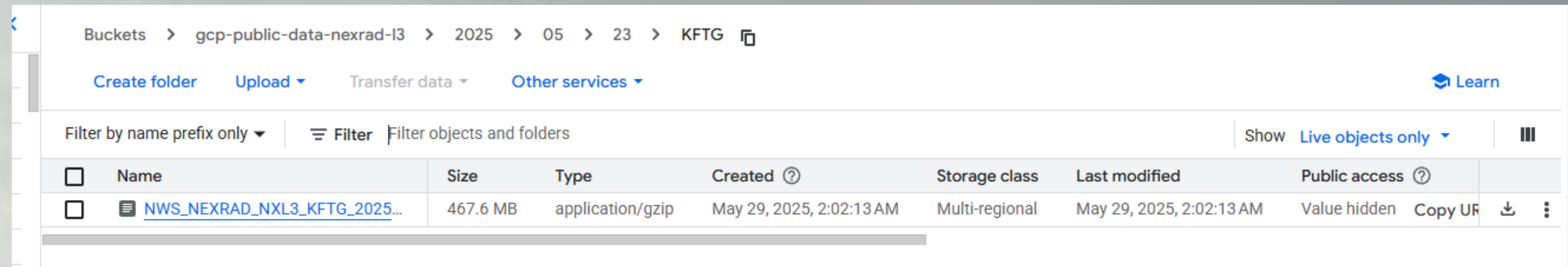
- Akron, Colorado, is in Washington County in NE CO, roughly 100mi NE of DEN
- In a favorable area for upslope flow and storm development
- Just southeast of the Cheyenne Ridge >> near a terrain feature that can influence storm development
- Akron in the 5% for tornado development on 05/23 at 1637Z



Less than a mile away!



DATA & METHODS

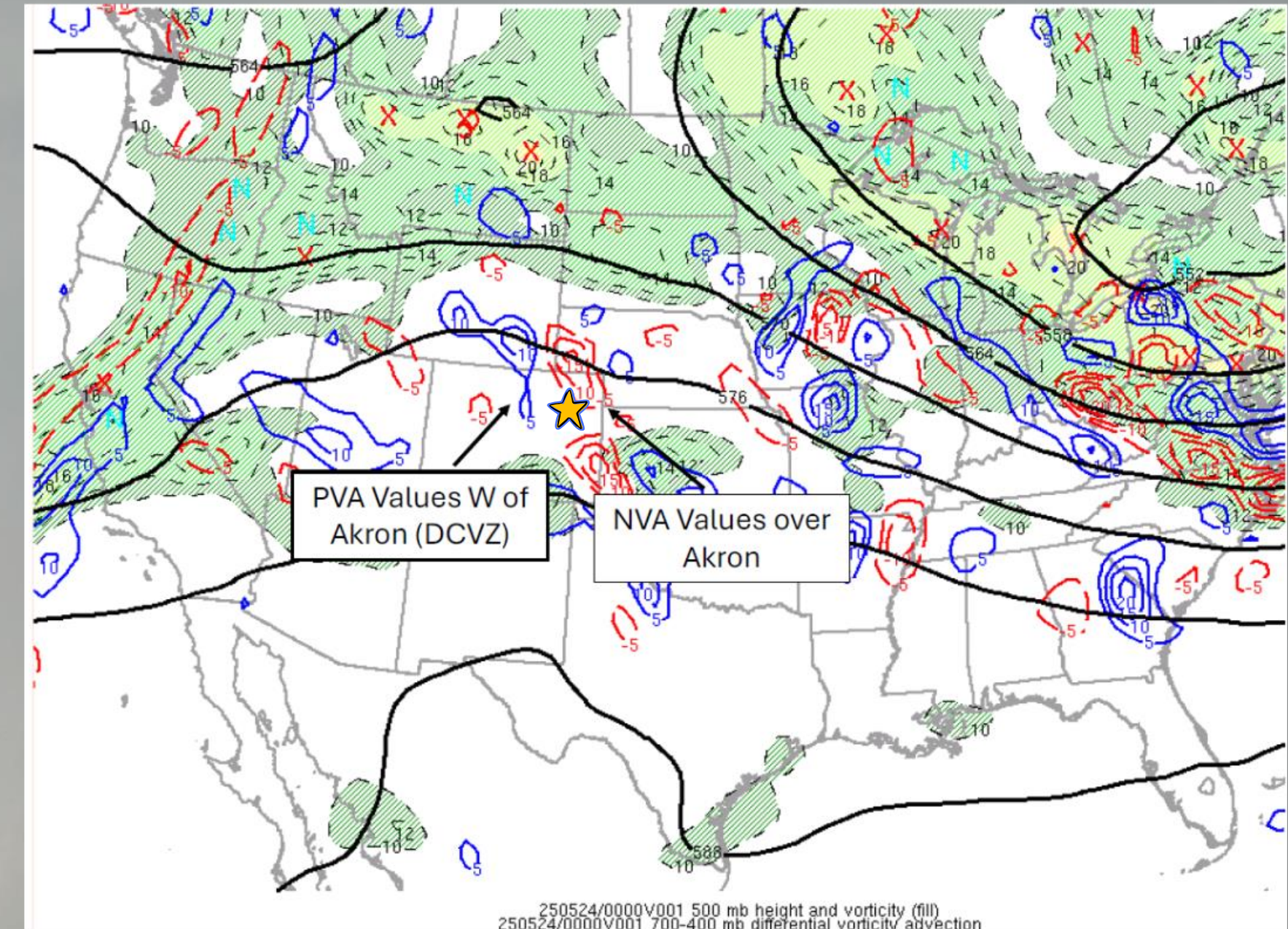
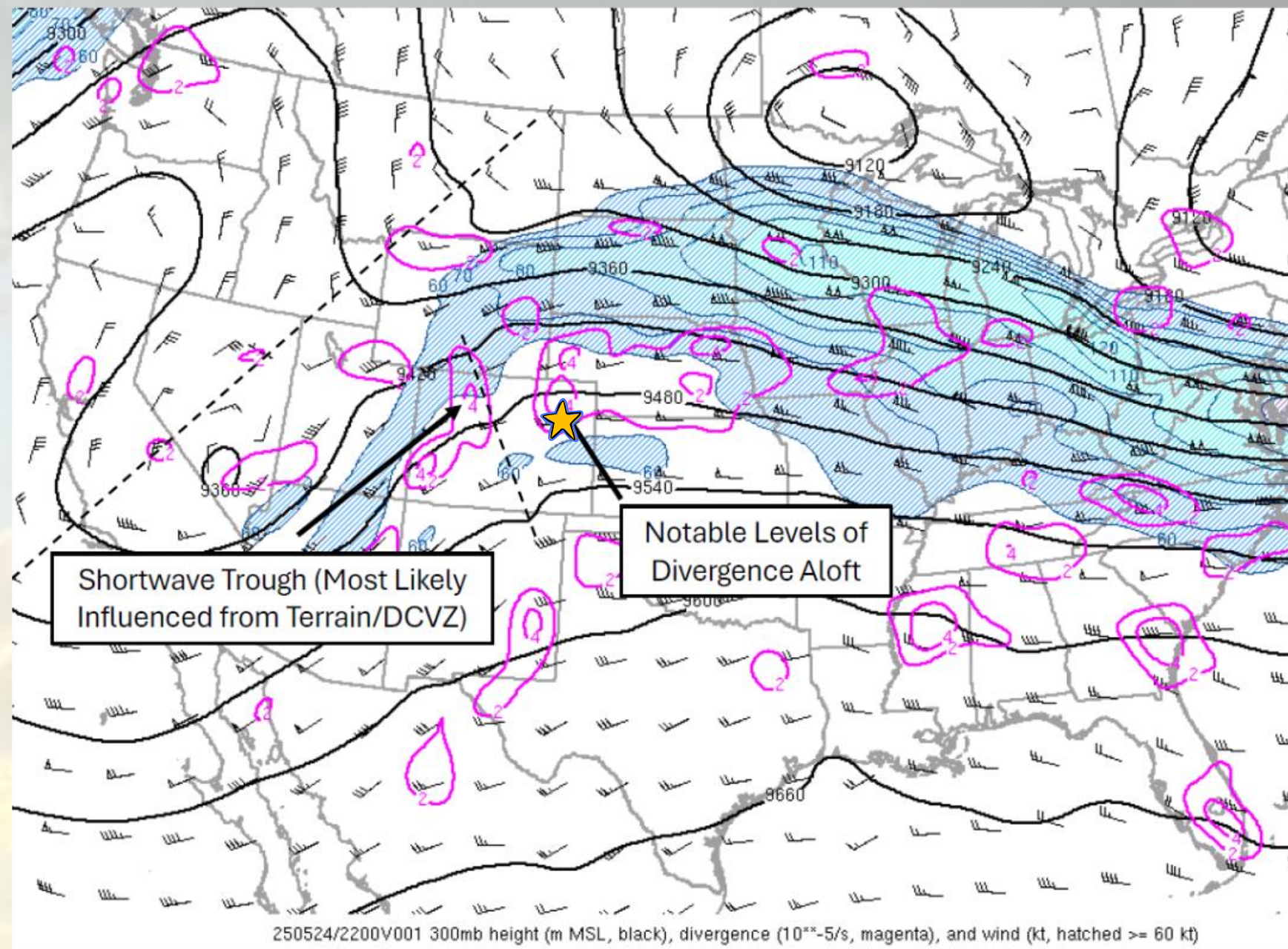


The screenshot shows the Google Cloud Storage interface. The breadcrumb path is Buckets > gcp-public-data-nexrad-l3 > 2025 > 05 > 23 > KFTG. The interface includes buttons for 'Create folder', 'Upload', 'Transfer data', and 'Other services'. A 'Filter' section is visible with 'Filter by name prefix only' and 'Filter objects and folders'. A table lists the objects in the bucket.

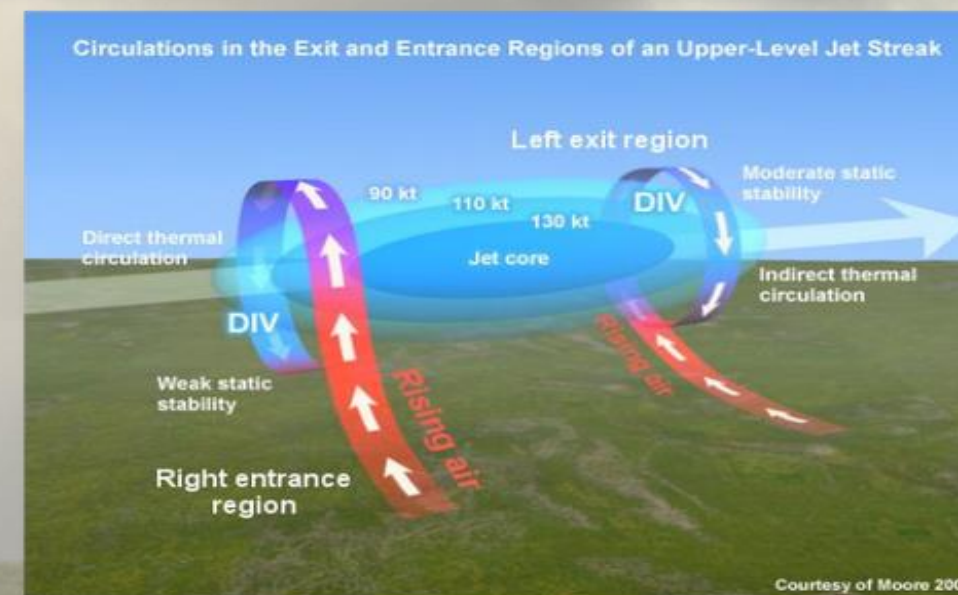
☐	Name	Size	Type	Created ?	Storage class	Last modified	Public access ?	
☐	 NWS_NEXRAD_NXL3_KFTG_2025...	467.6 MB	application/gzip	May 29, 2025, 2:02:13 AM	Multi-regional	May 29, 2025, 2:02:13 AM	Value hidden	Copy UR  

- Satellite Data: Retrieved from **COD Satellite Archive**; Times pulled were **2157z-2357z on 05/23/2025**.
- Radar Data: Retrieved from the **Google Cloud Network** and **Amazon AWS** for Level II and Level III Data. Observed times were identical to satellite data.
- Synoptic/Mesoscale parameters were observed from 1200z on May 23rd to 0600z on May 24th to determine forcing mechanisms and influences.
- Satellite/Radar allowed for a real-time analysis of the storm's life cycle.

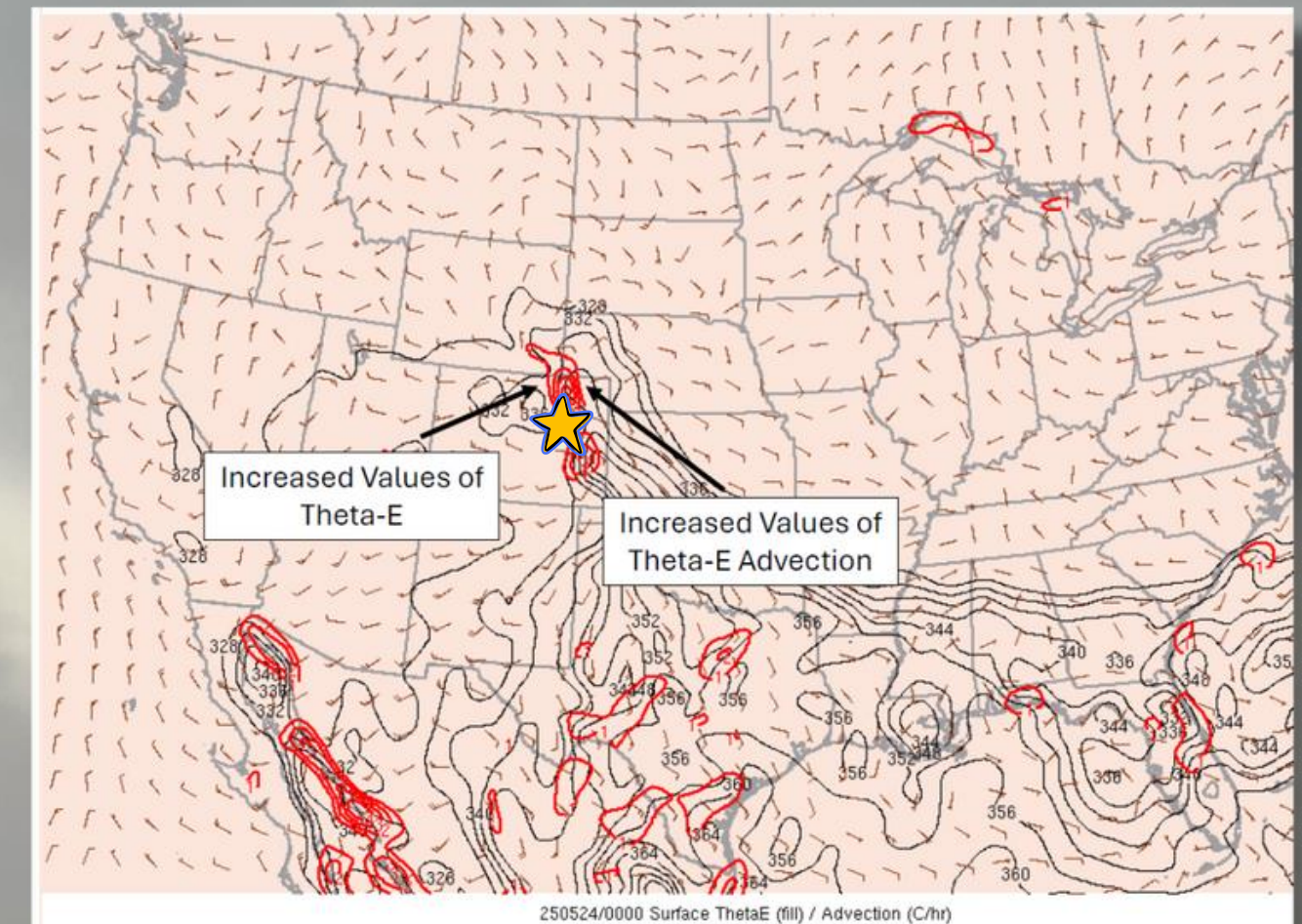
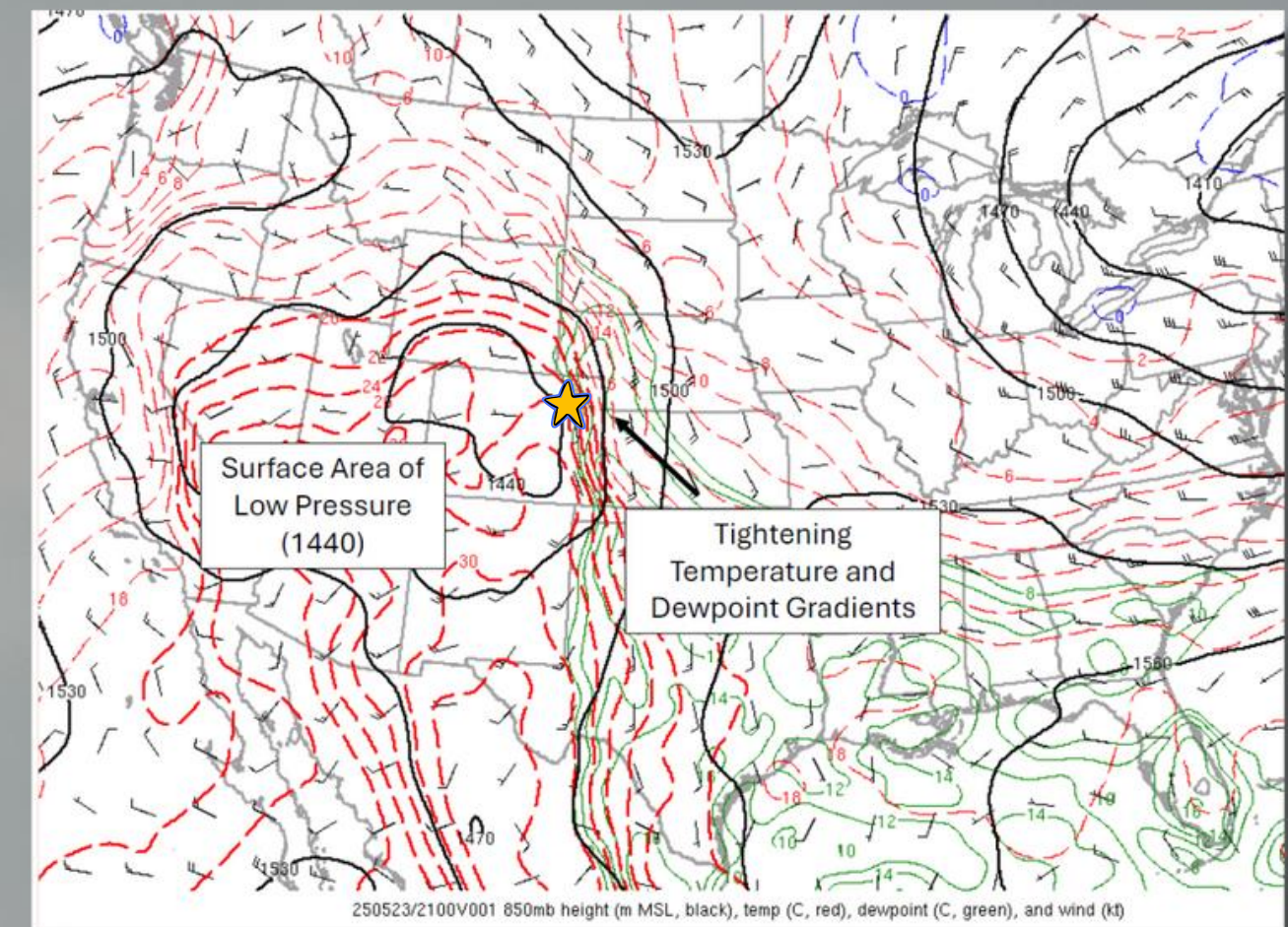
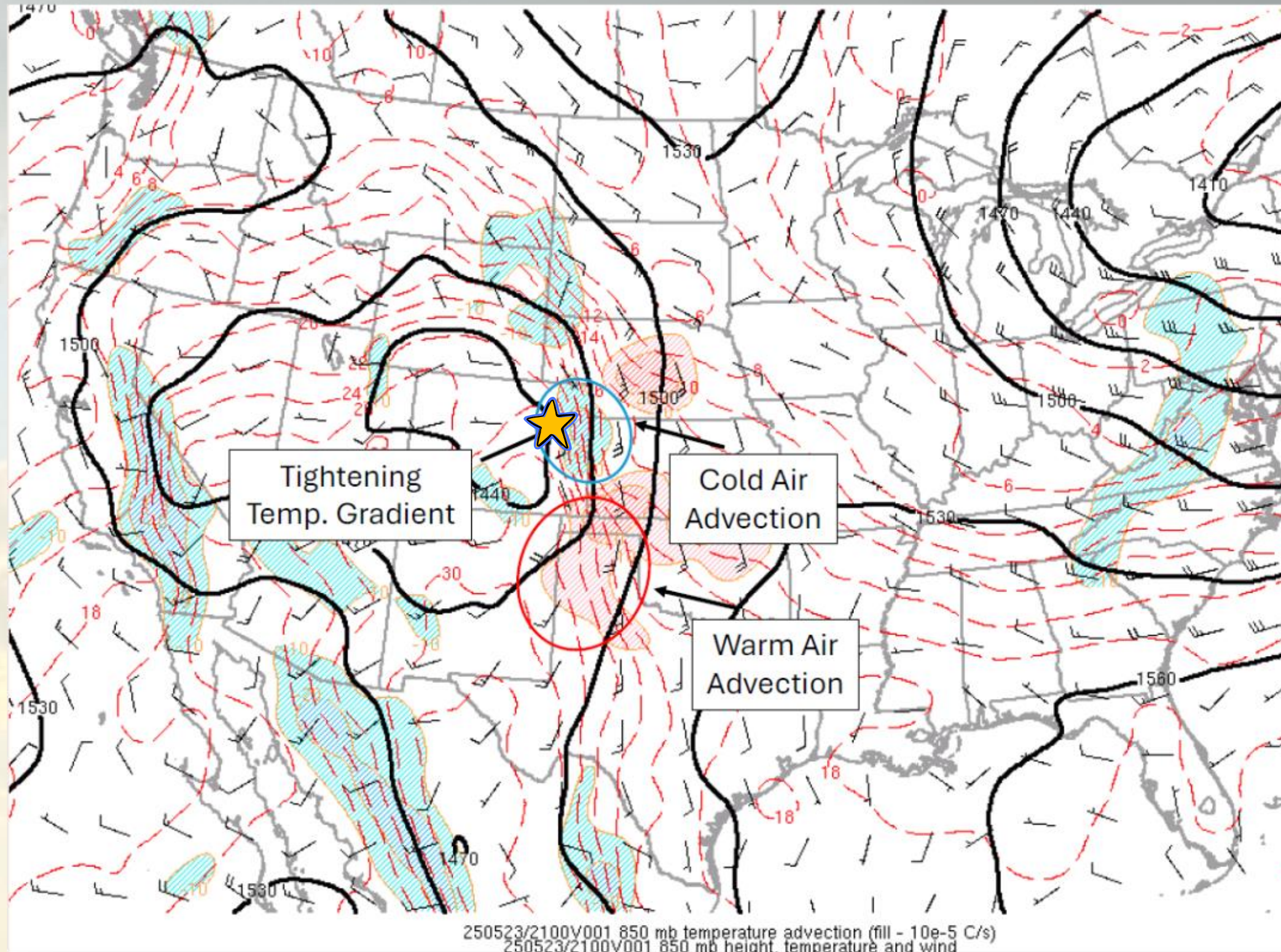
SYNOPTIC SETUP



- Akron located in the right entrance region of the jet streak
- Weak but sufficient upper-level div.
- Small height falls support small ascent

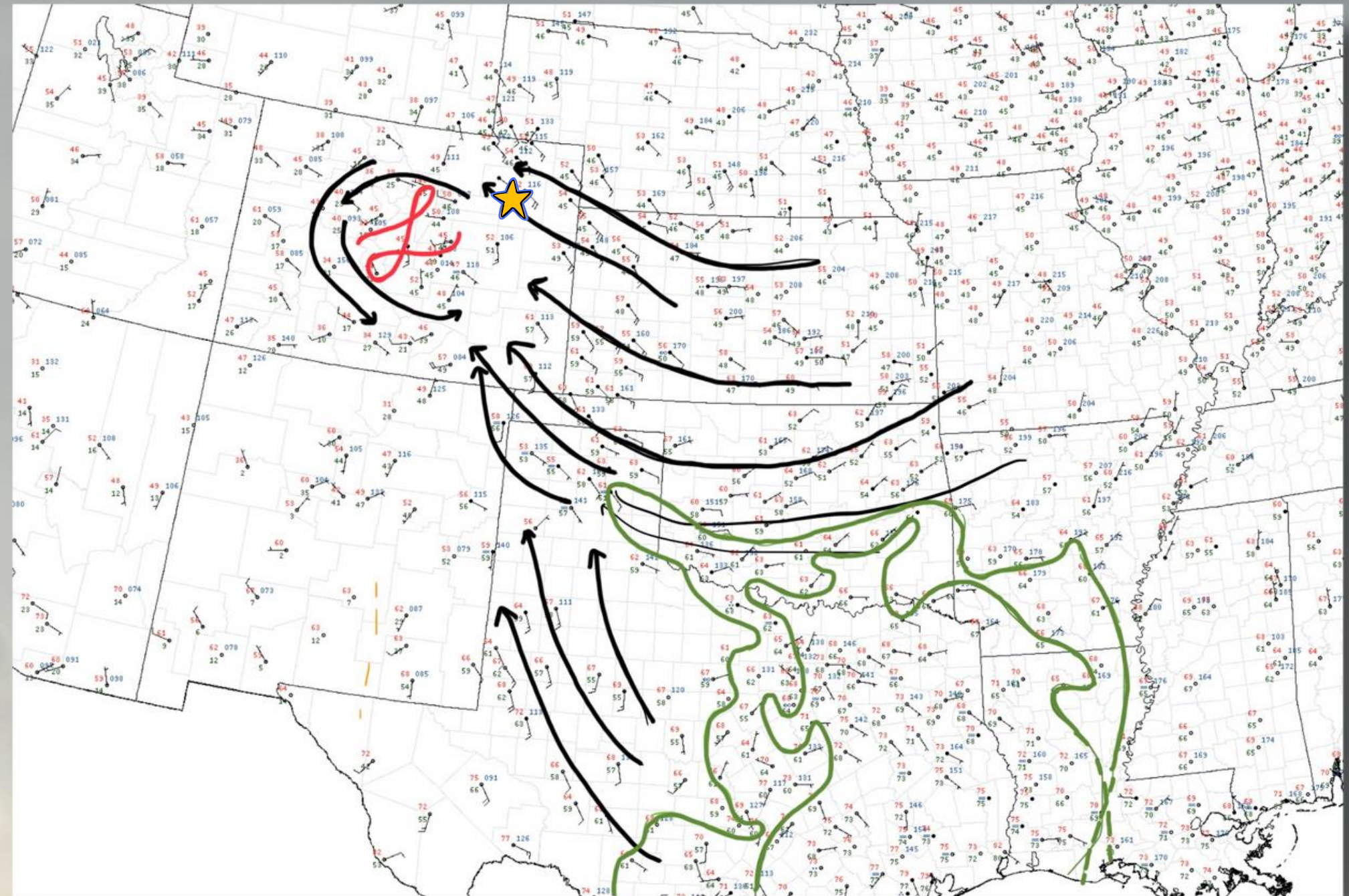
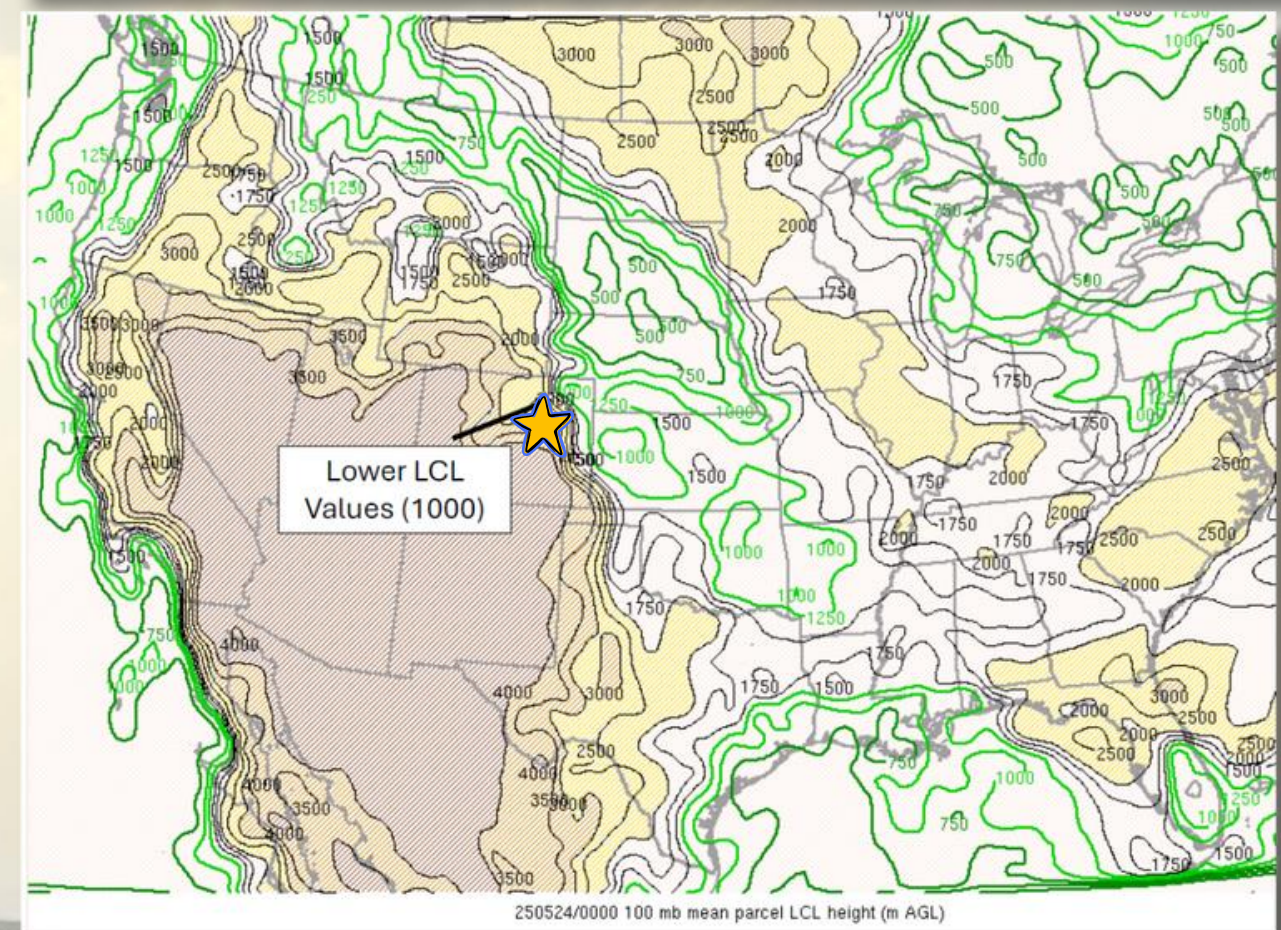
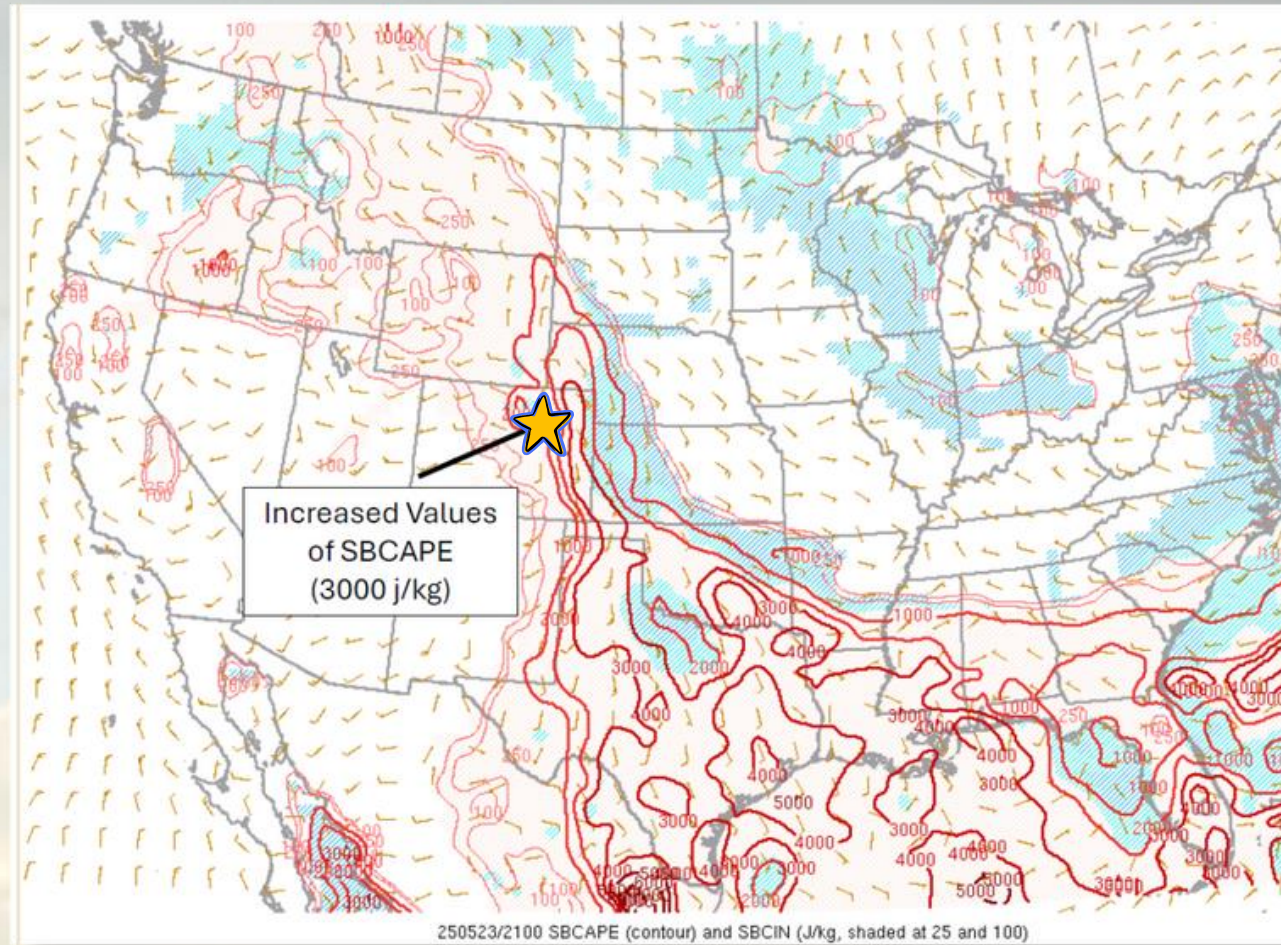


SYNOPTIC SETUP



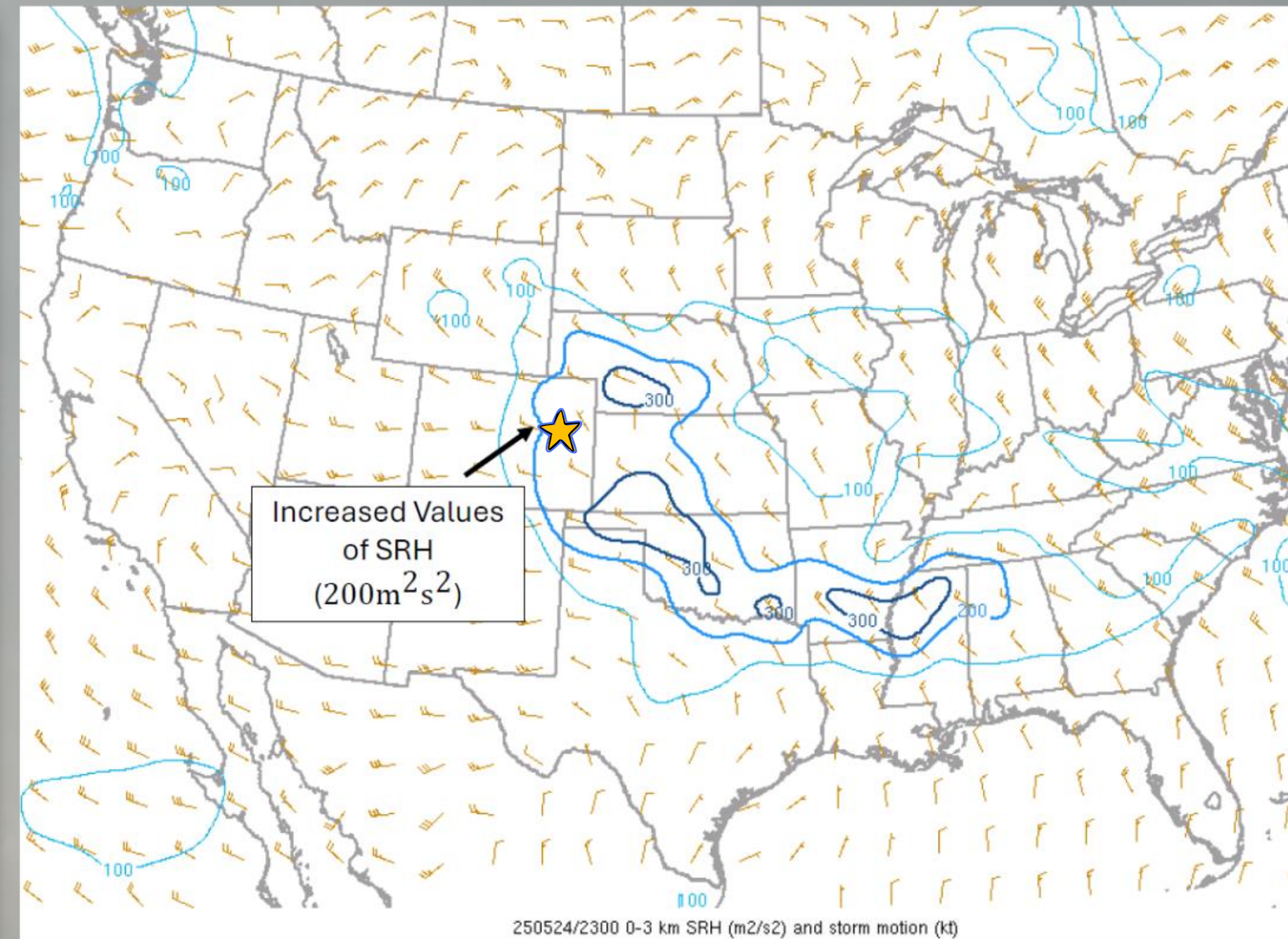
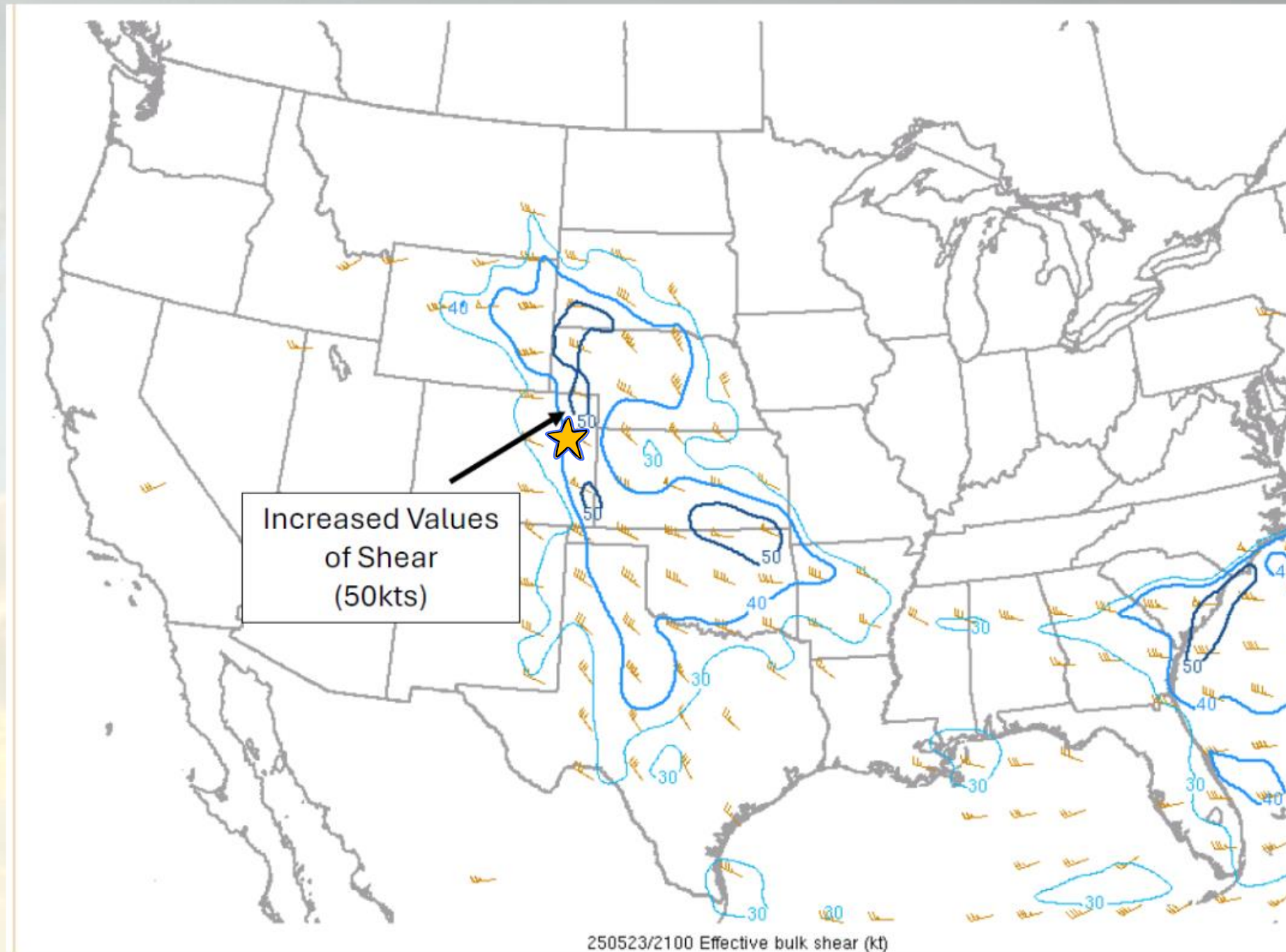
- Tightening 850-mb temperature & dewpoint gradients = increased baroclinicity and low-level convergence
- Sharpening θ_e gradient $\gg$ zone of strongest instability and moisture
- Enhanced thermal/ θ_e contrast $\gg$ low-level vorticity and boundary forcing for initiation

MESO SETUP



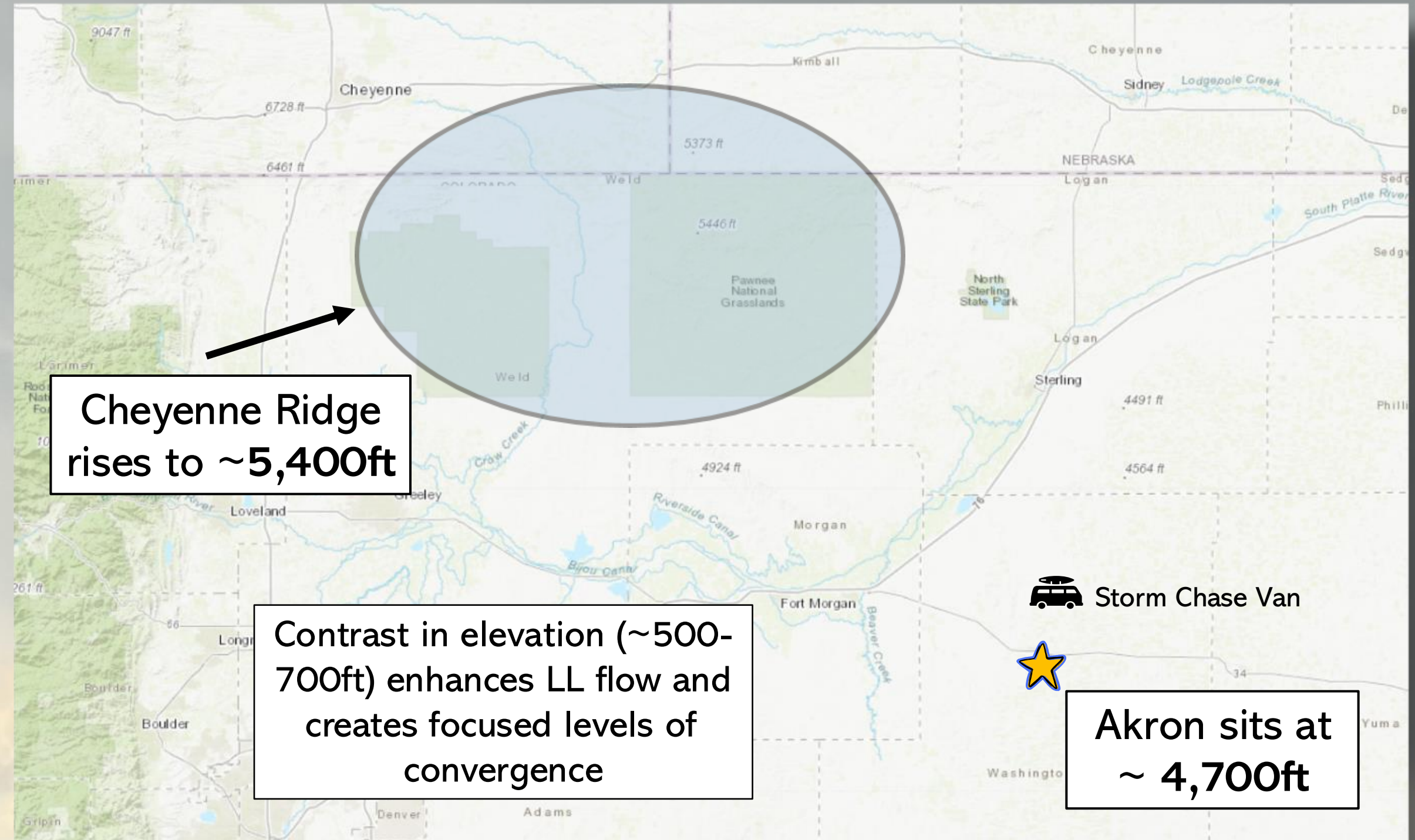
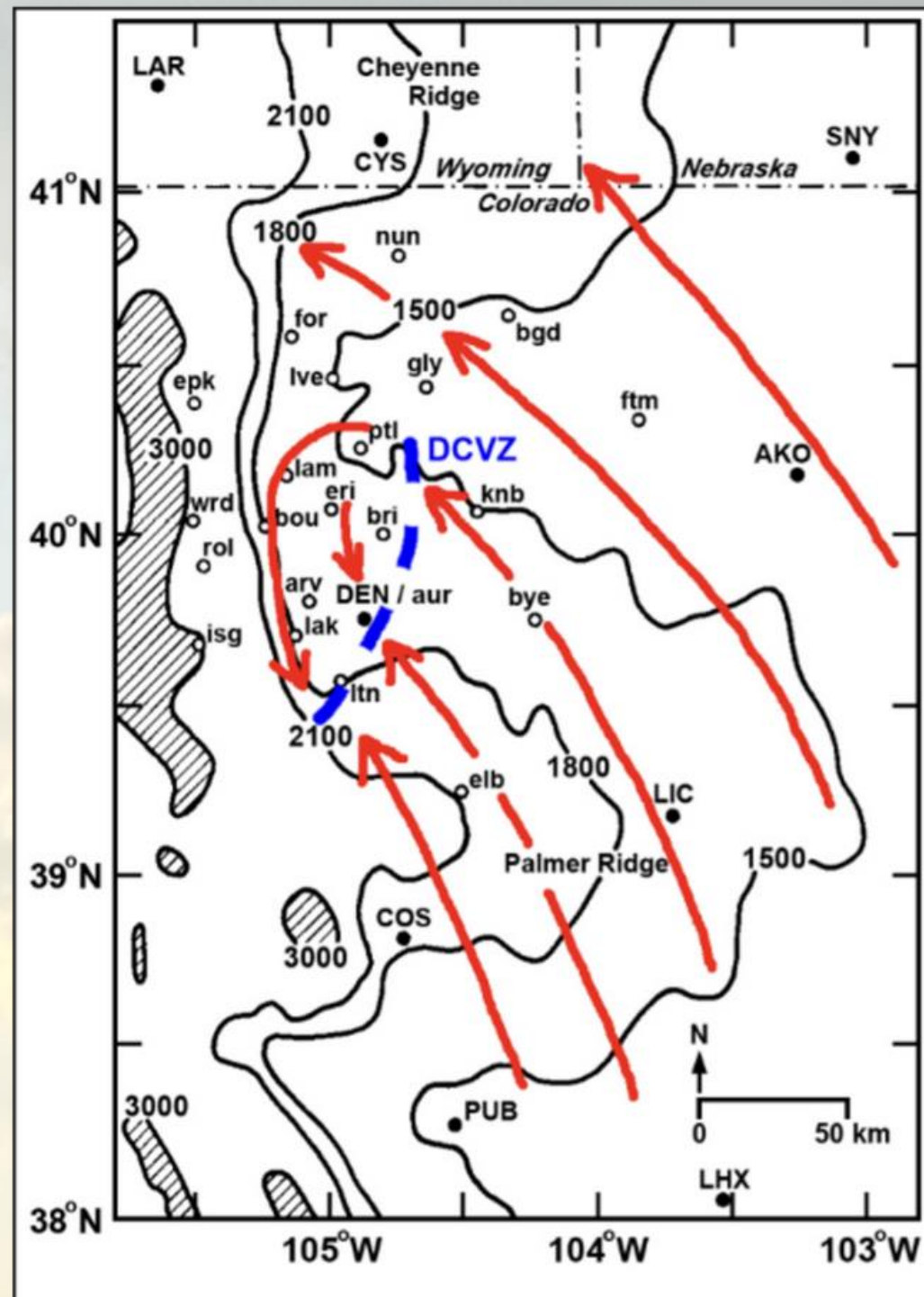
- Higher instability (SBCAPE) values centered over Akron
- Lower LCL levels = easier for vorticity to stretch into the vertical
- Early in the day, higher dewpoint values begin nosing into SFC low in Akron

MESO SETUP



- Favorable shear values supported supercell structure
- Enhanced areas of SRH over Akron

TERRAIN INFLUENCE AT THE CHEYENNE RIDGE



- Terrain-induced convergence increases lift near the SFC
- Upslope flow from the ridge
- Local modification of LL wind profile

SATELLITE & RADAR DATA

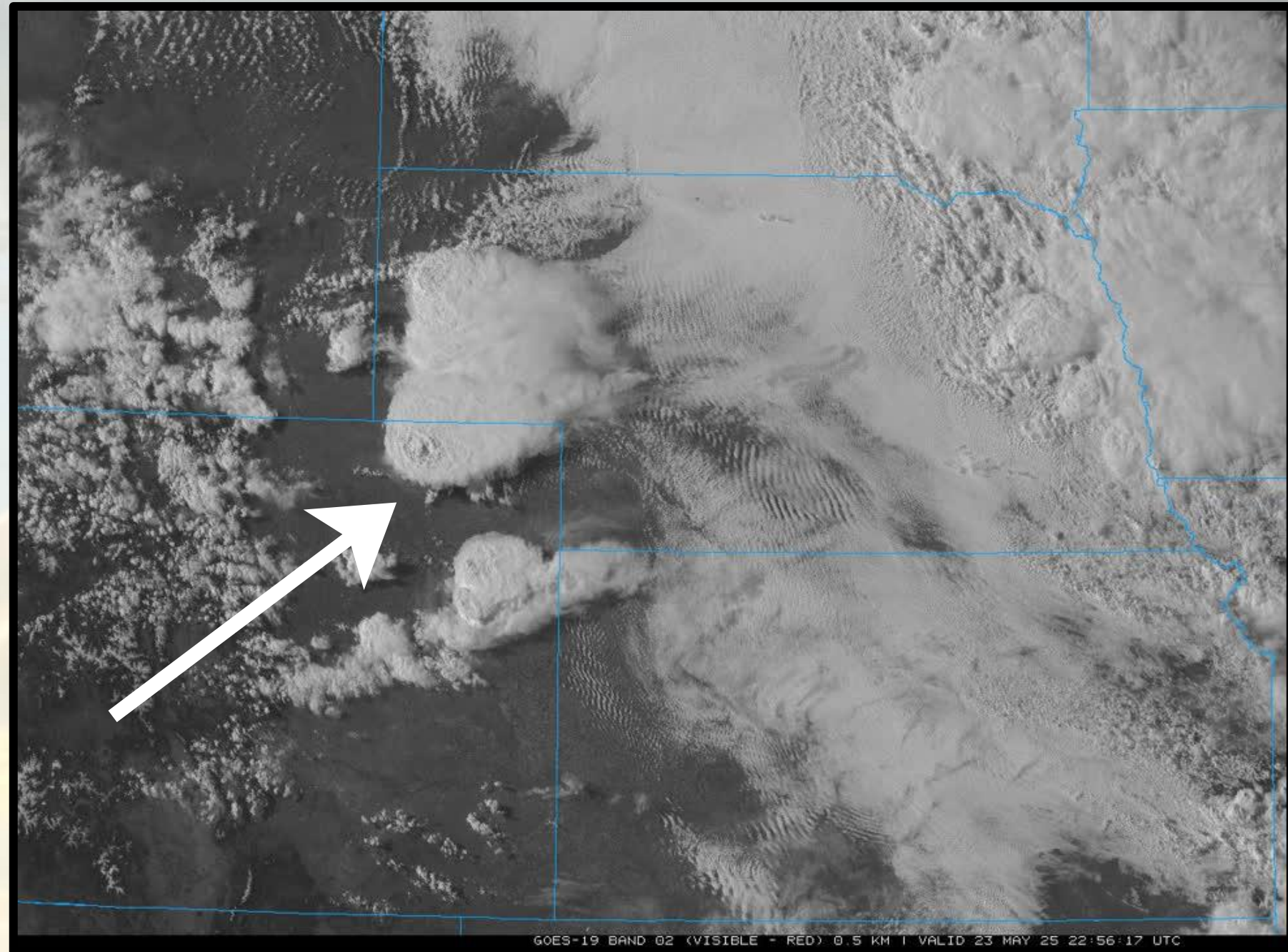
SAT:

1. Visible Blue (ABI 1)
2. Lower-Level Water Vapor (ABI 10)
3. Clean Longwave Infrared (ABI 13)

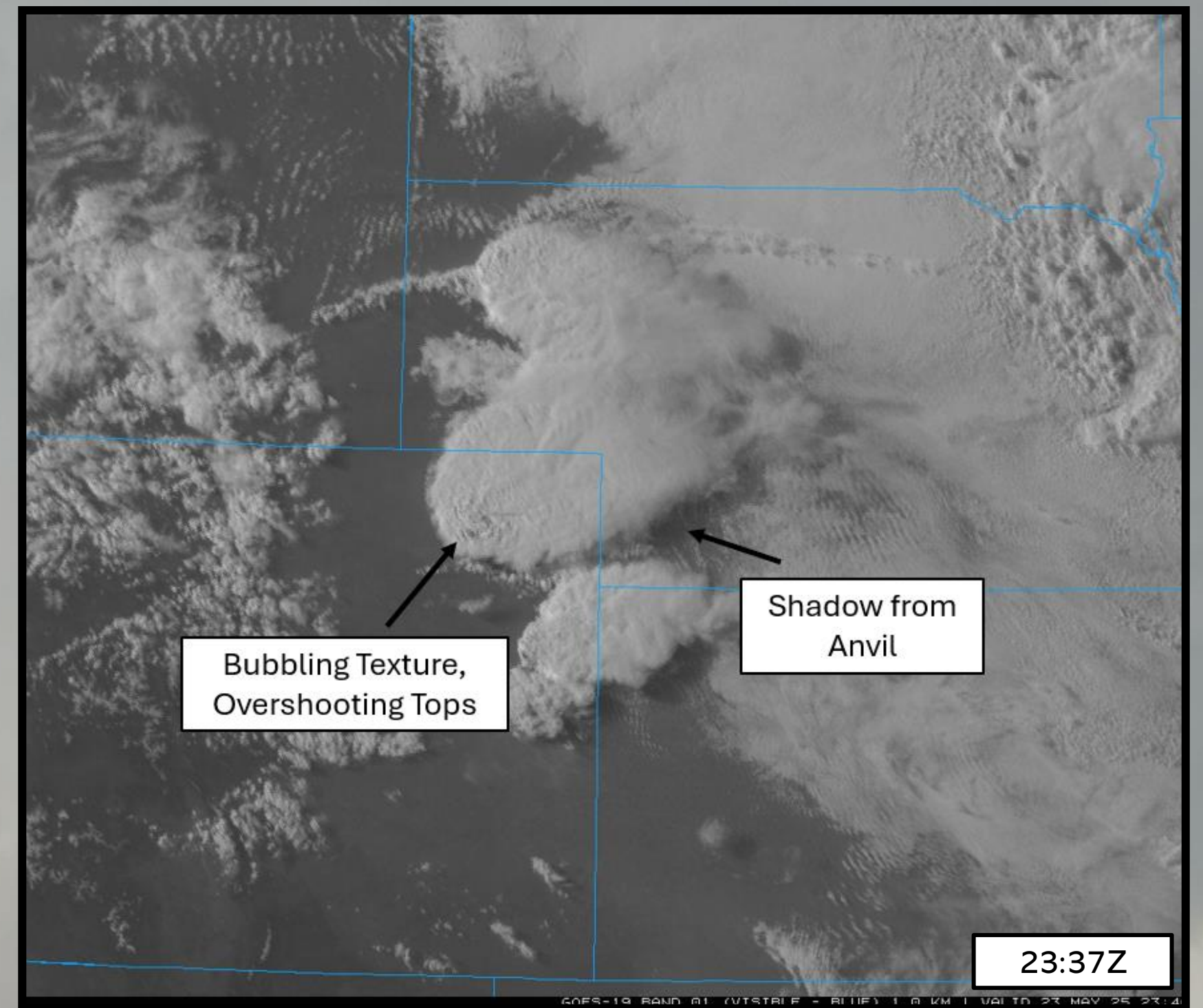
RDR:

- | | |
|----------------------------|--------------------------------|
| 1. Base Reflectivity | 6. Correlation Coefficient |
| 2. Base Velocity | 7. Specific Differential Phase |
| 3. Storm Relative Velocity | 8. POSH |
| 4. Spectrum Width | 9. VIL |

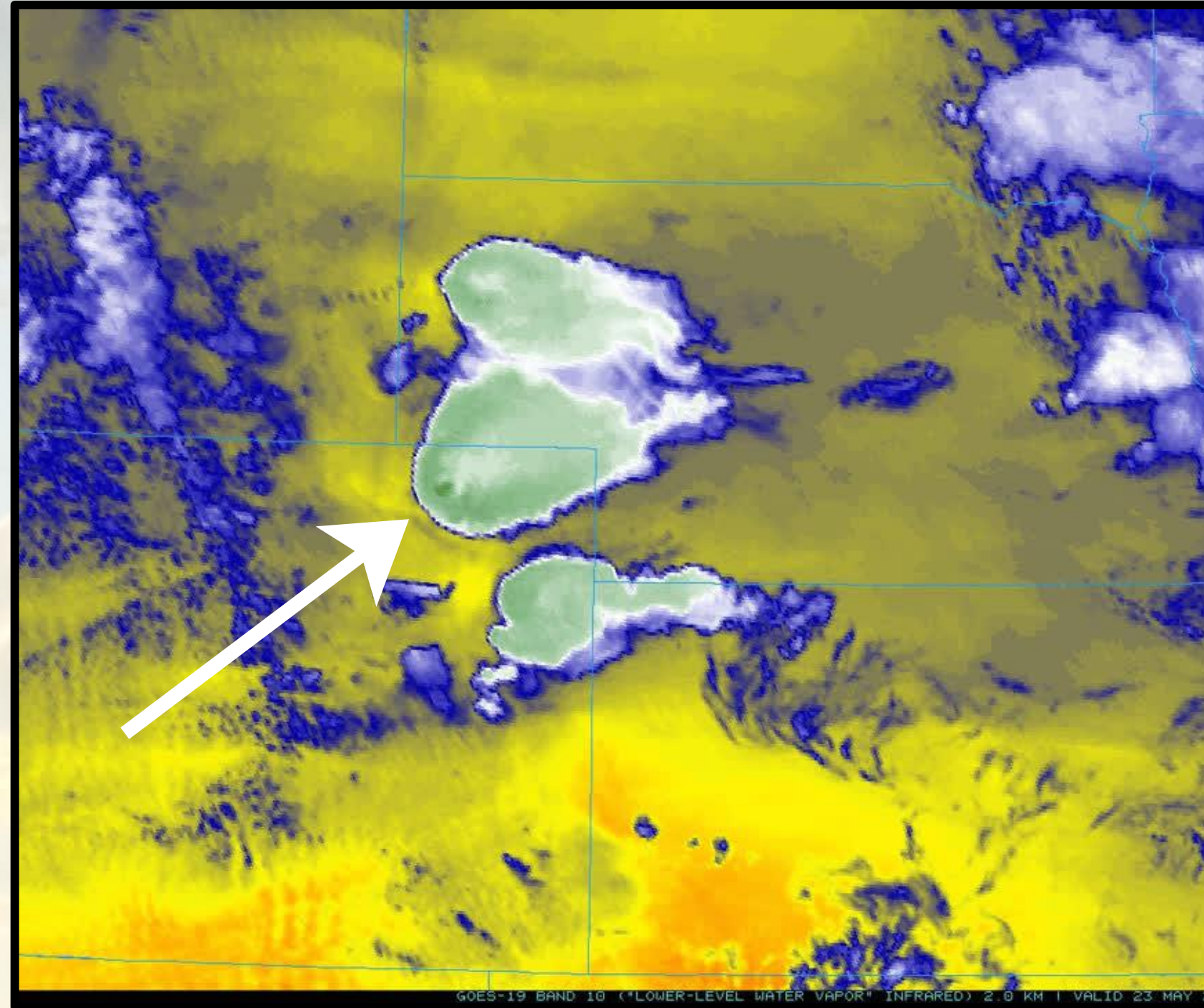
SATELLITE – VISIBLE BLUE (ABI 1)



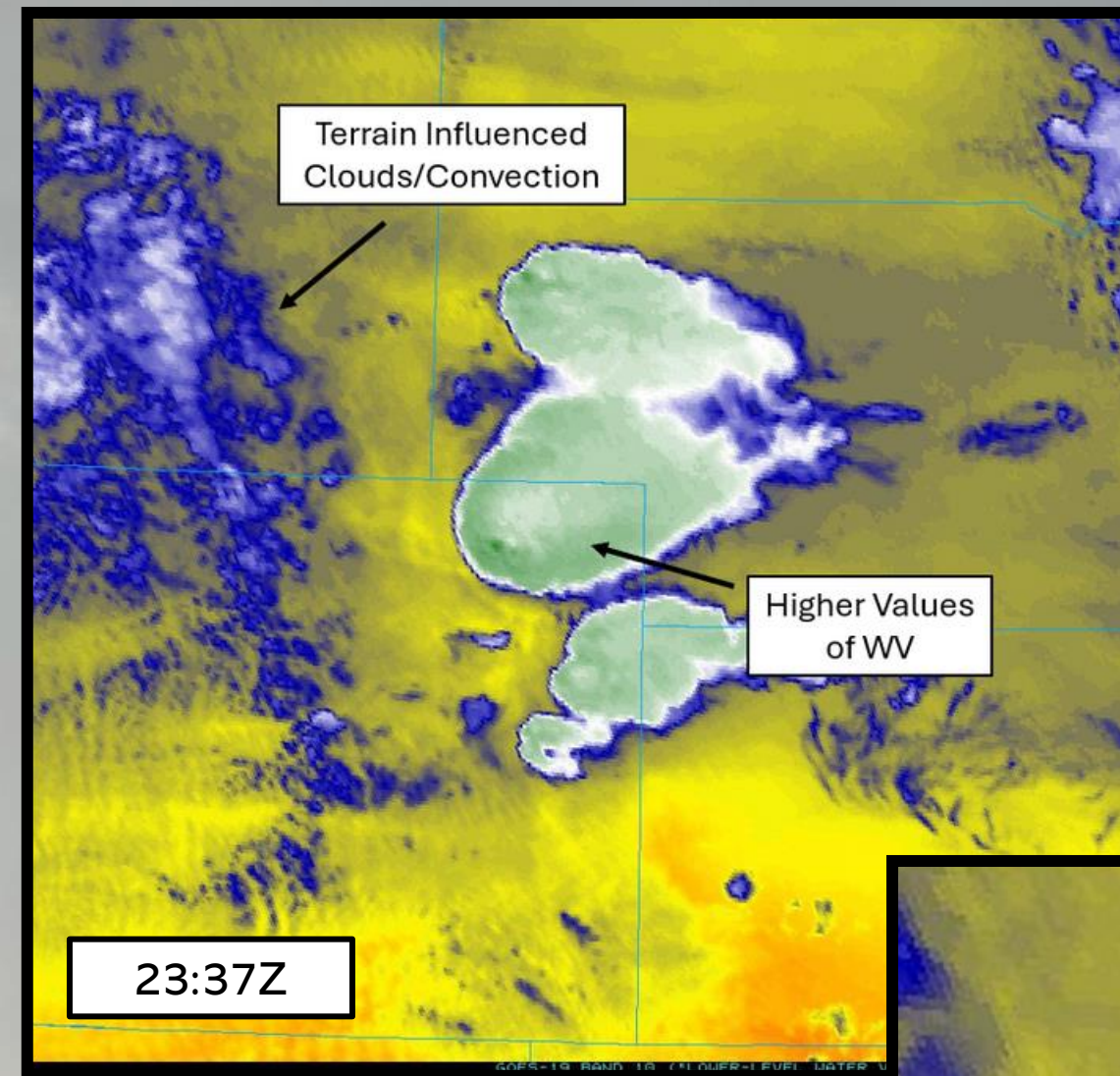
- Overshooting tops on the supercell of interest
- Bubbling texture, indicative of strong updrafts
- Above Anvil Cirrus Plume (AACP)



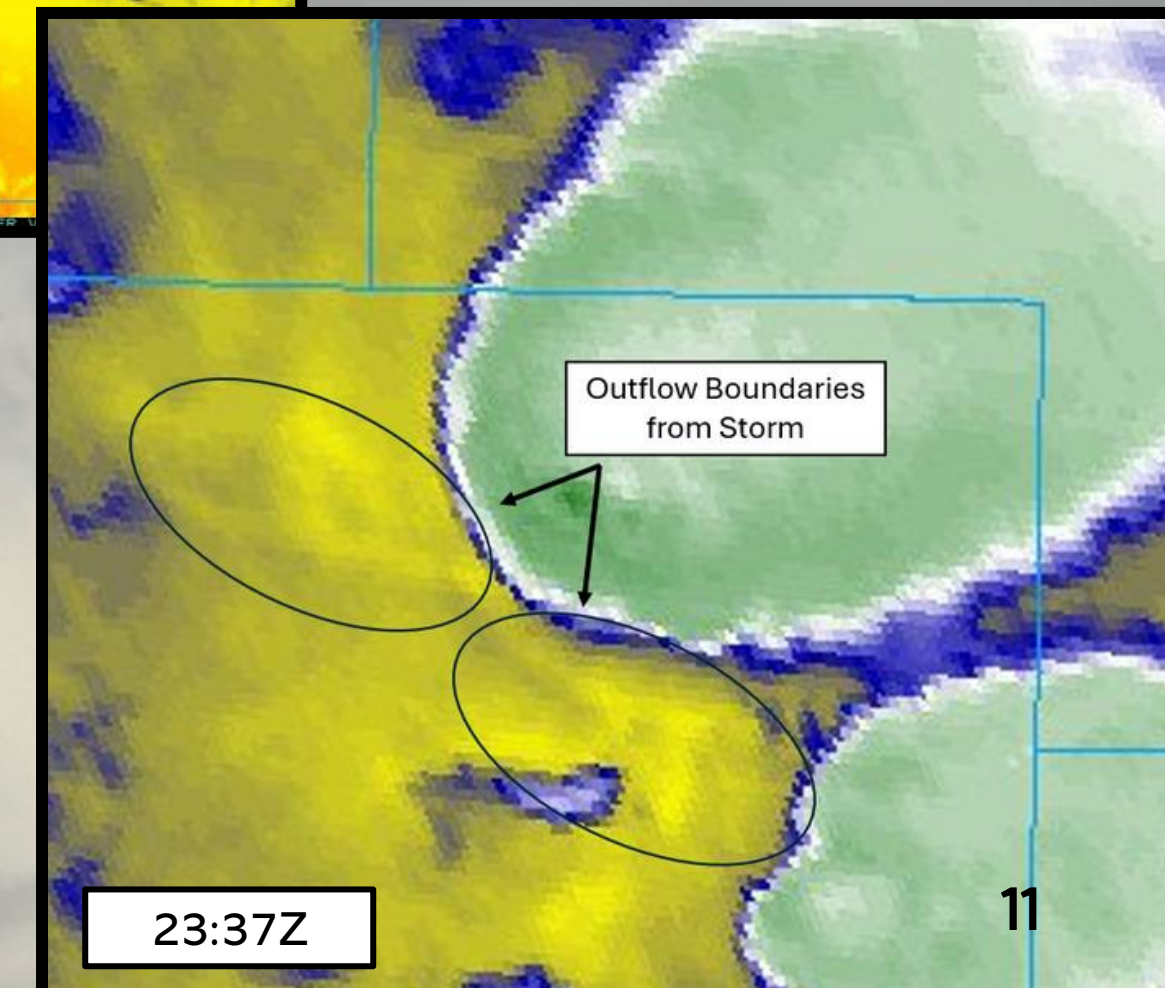
SATELLITE – LL WATER VAPOR (ABI 10)



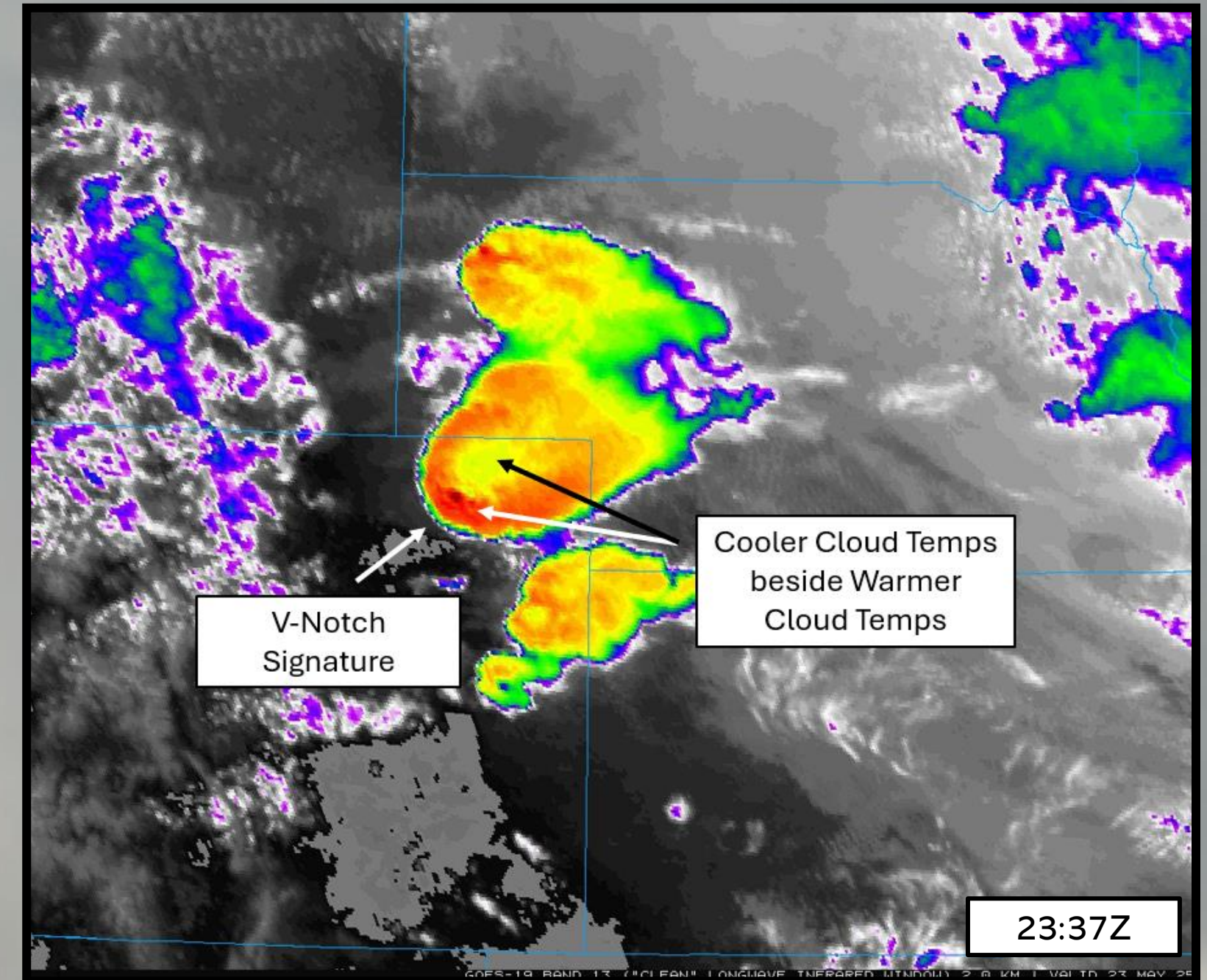
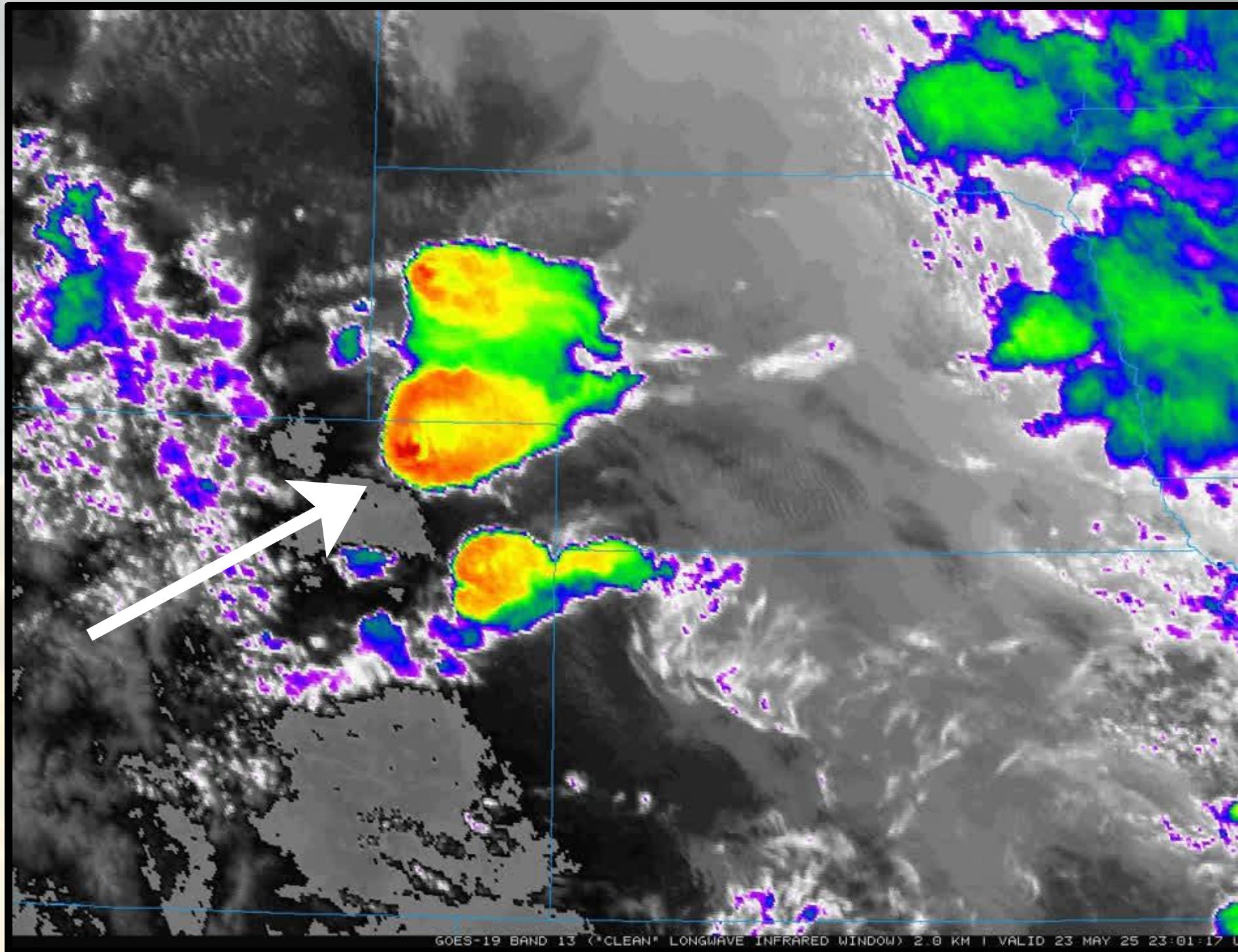
- Higher values of WV = moisture into storm
- Dark green color over updraft area = efficient transport of moisture



- OFBs show cooler air spreading outward
- Increase in LL convergence, can set off new convection

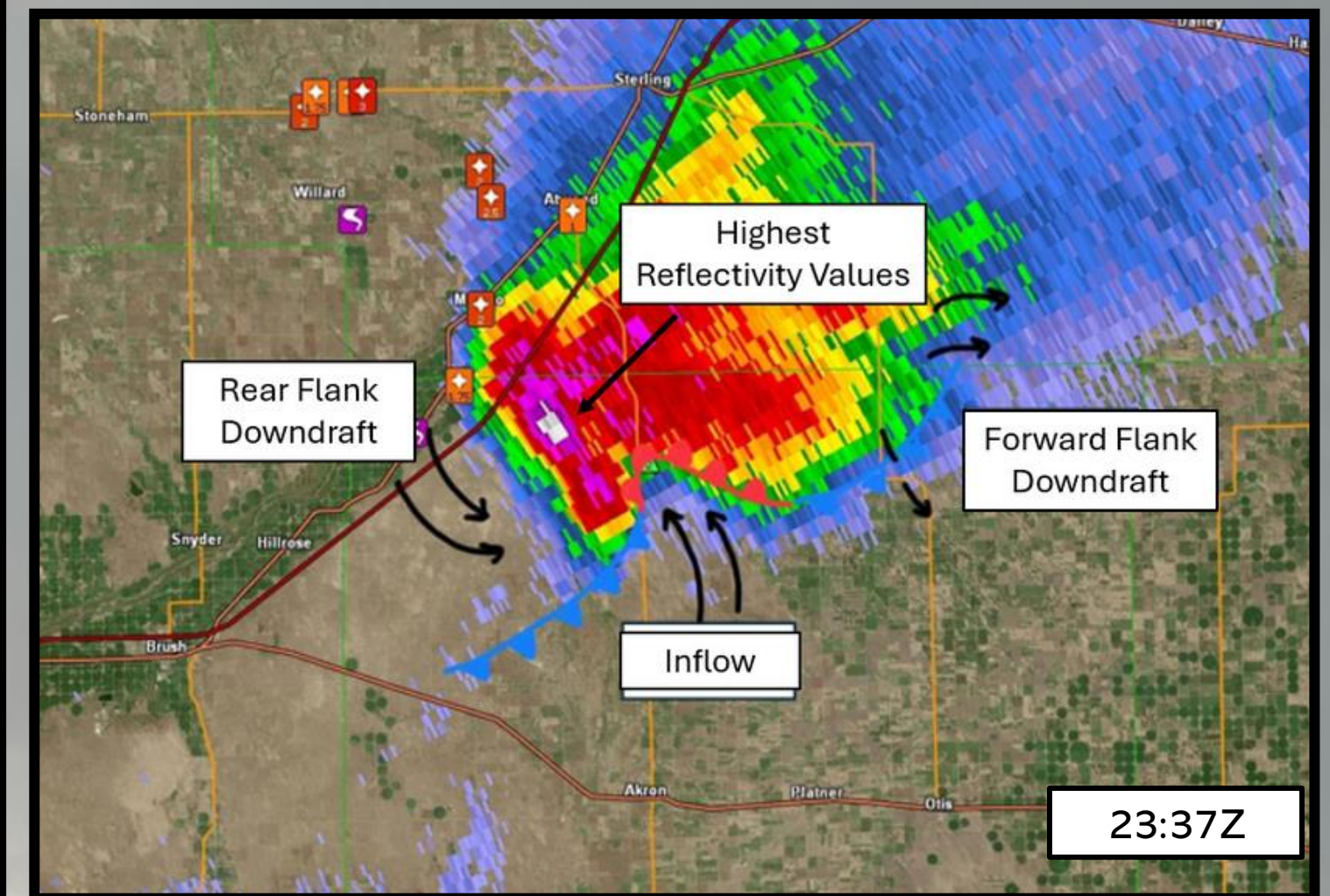
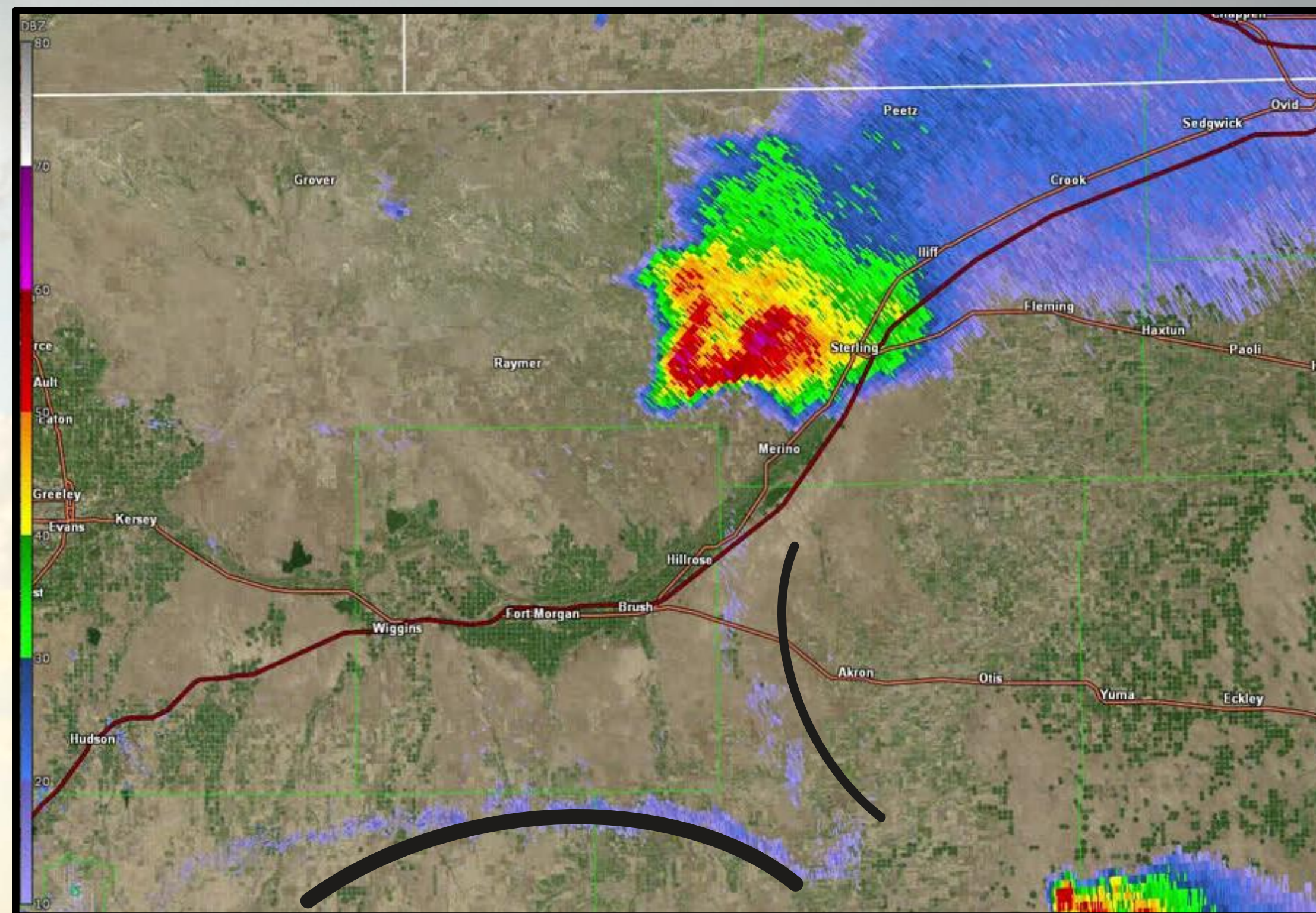


SATELLITE – LONGWAVE IR (CLEAN) (ABI 13)



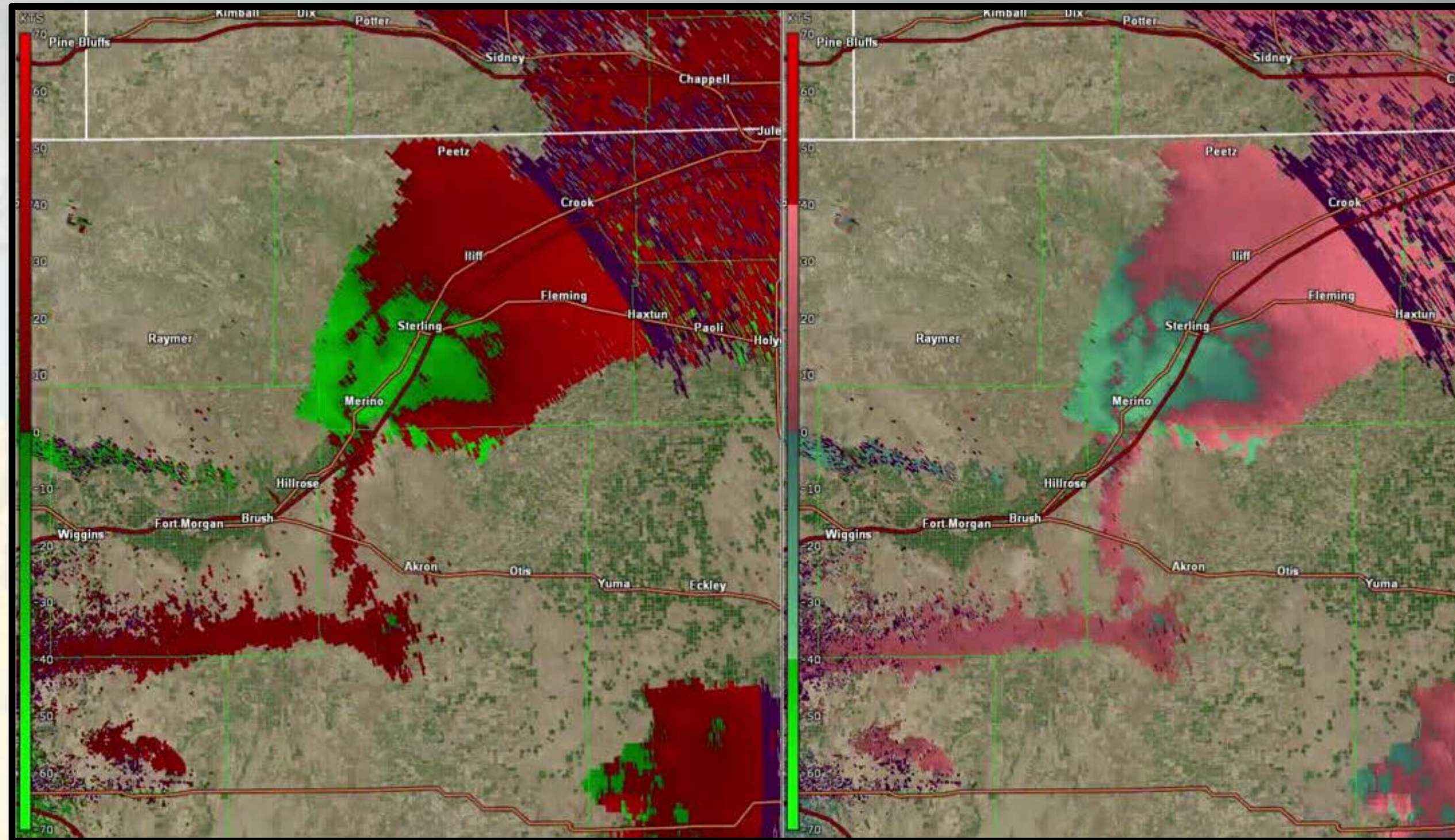
- Overshooting tops on the supercell of interest
- Cold cloud tops indicative of a strong updraft

RADAR – BASE REFLECTIVITY

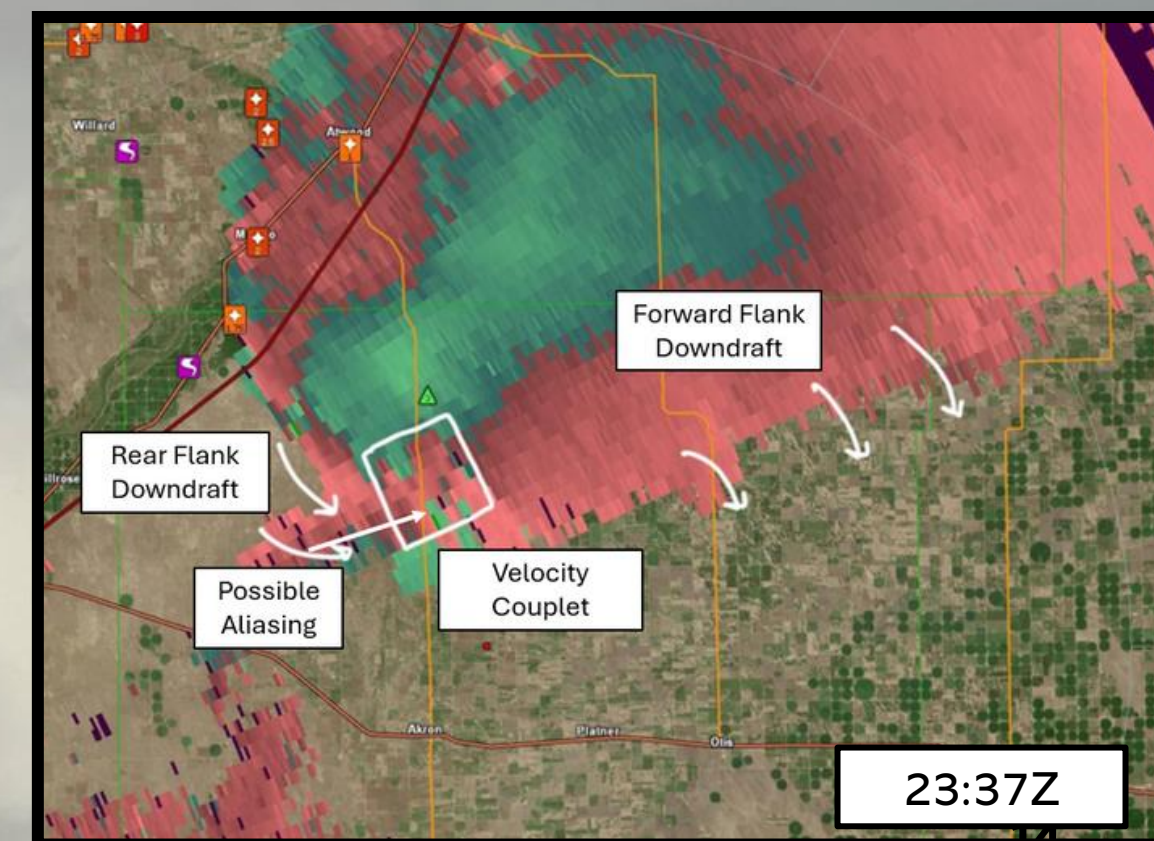
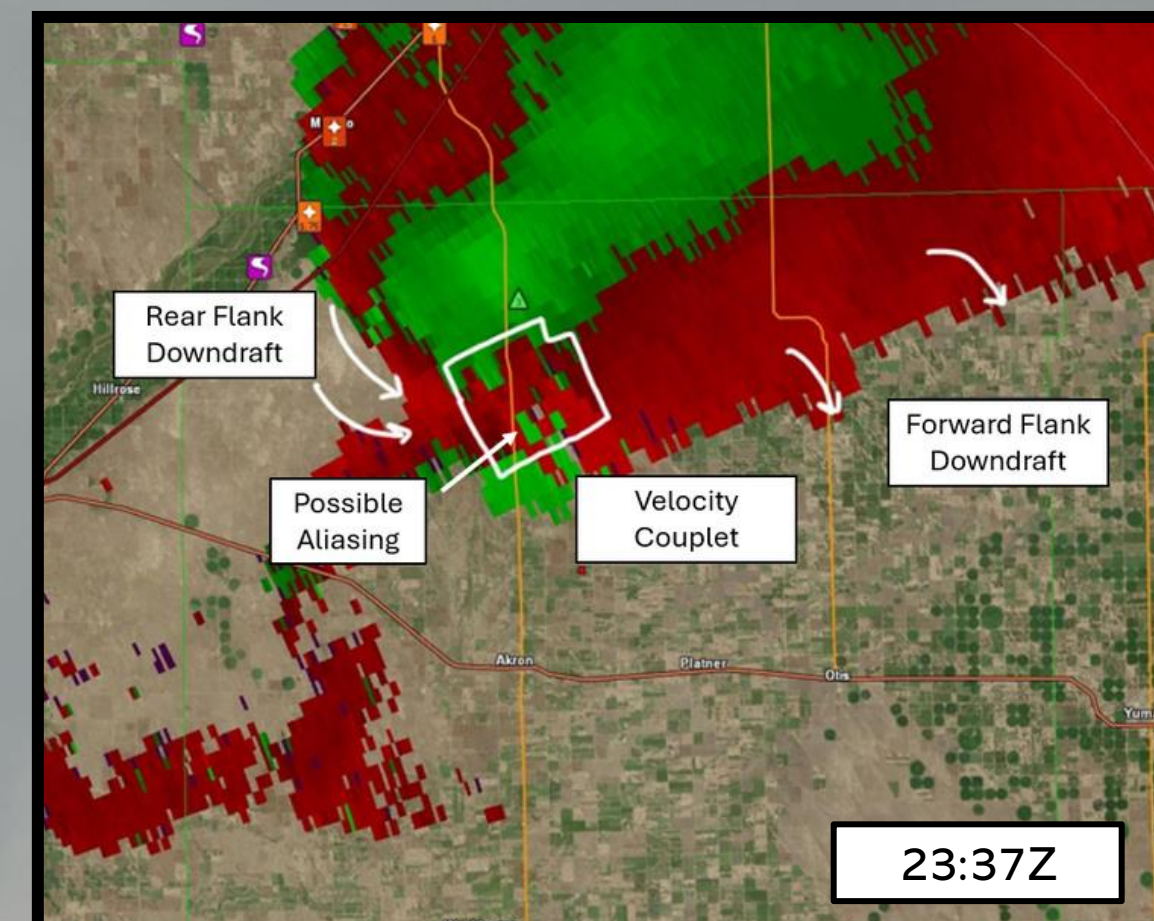


- Storm latches onto the boundary as it moves southeast
- Strong area of increased reflectivity values

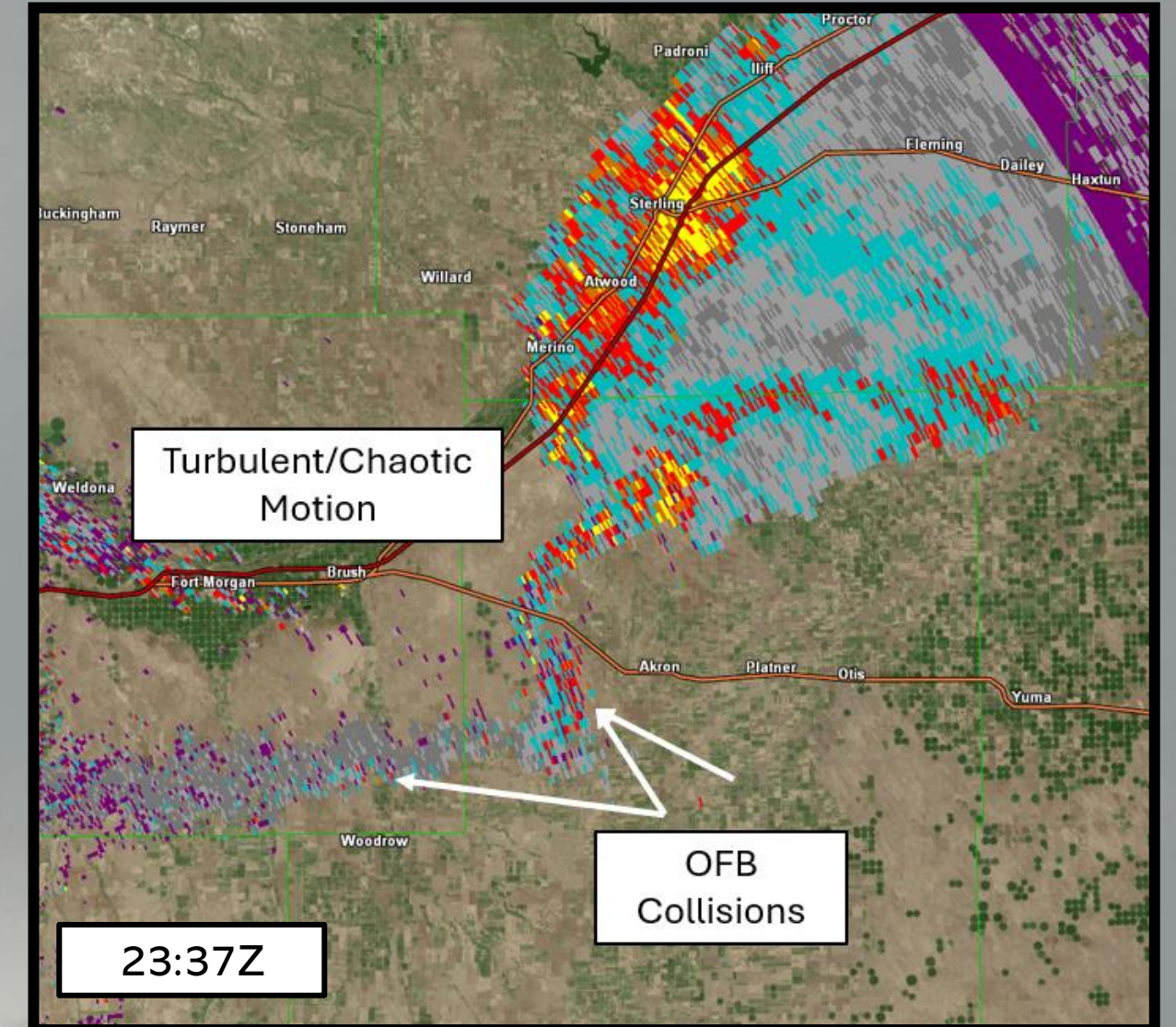
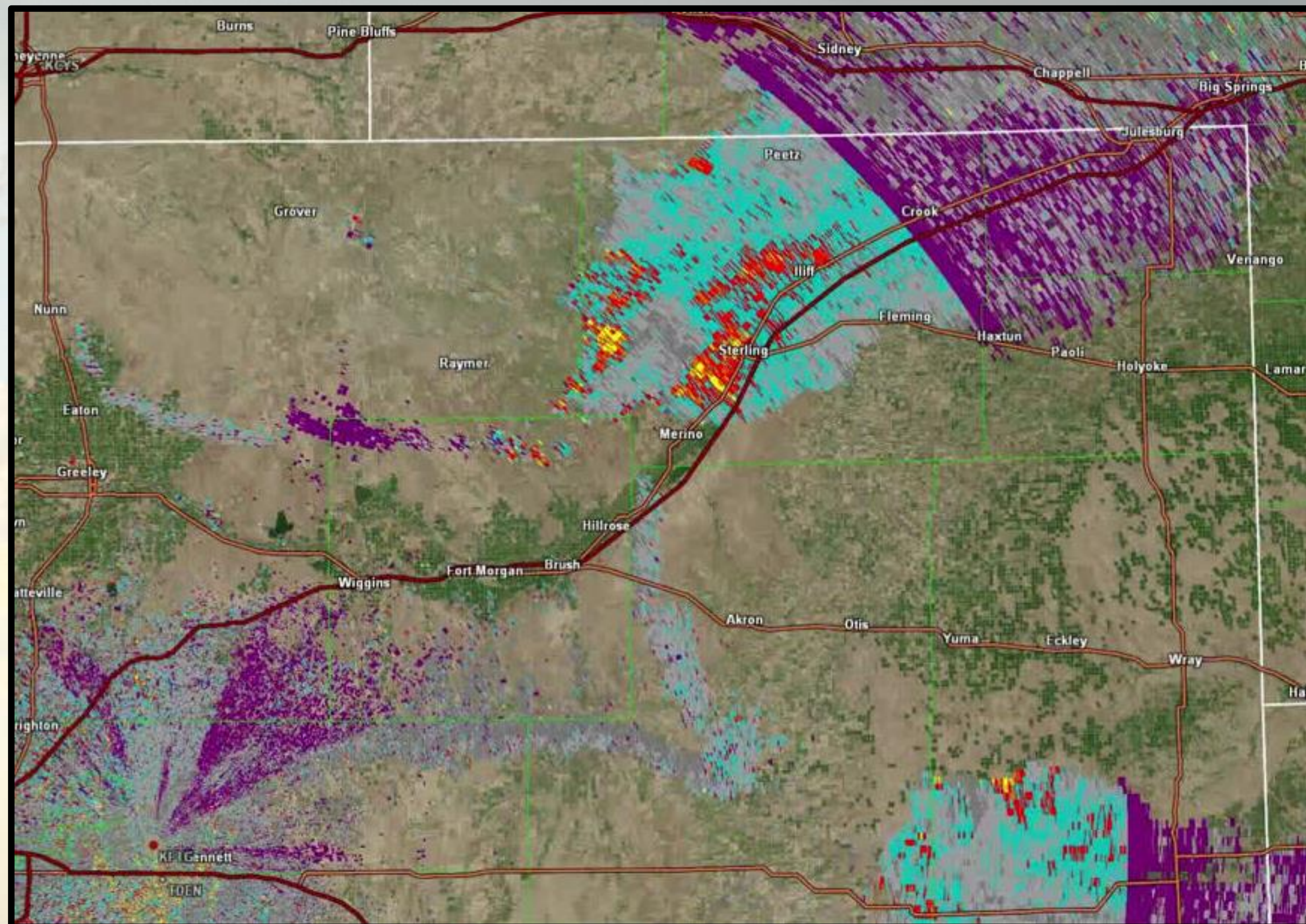
RADAR – BASE & STORM REL. VELOCITY



- Storm latches onto the boundary as it is moving southeast
- Possible aliasing in both Base and Storm Relative Velo.

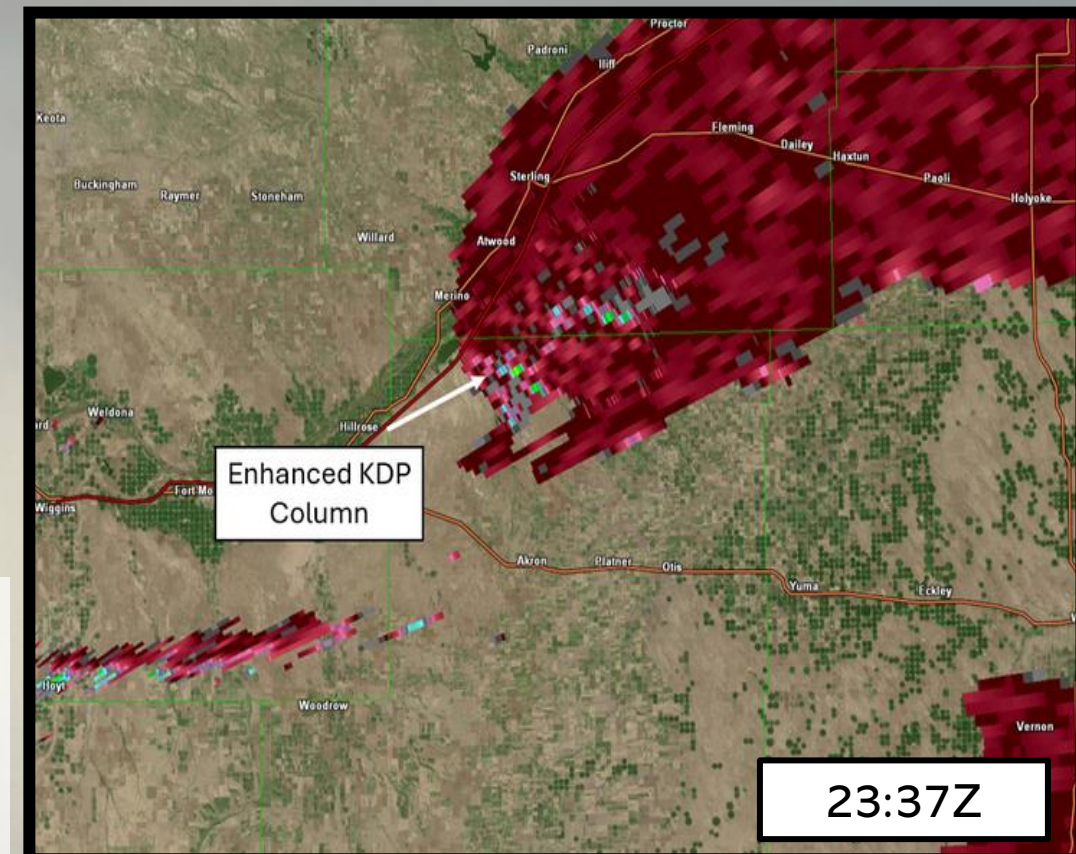
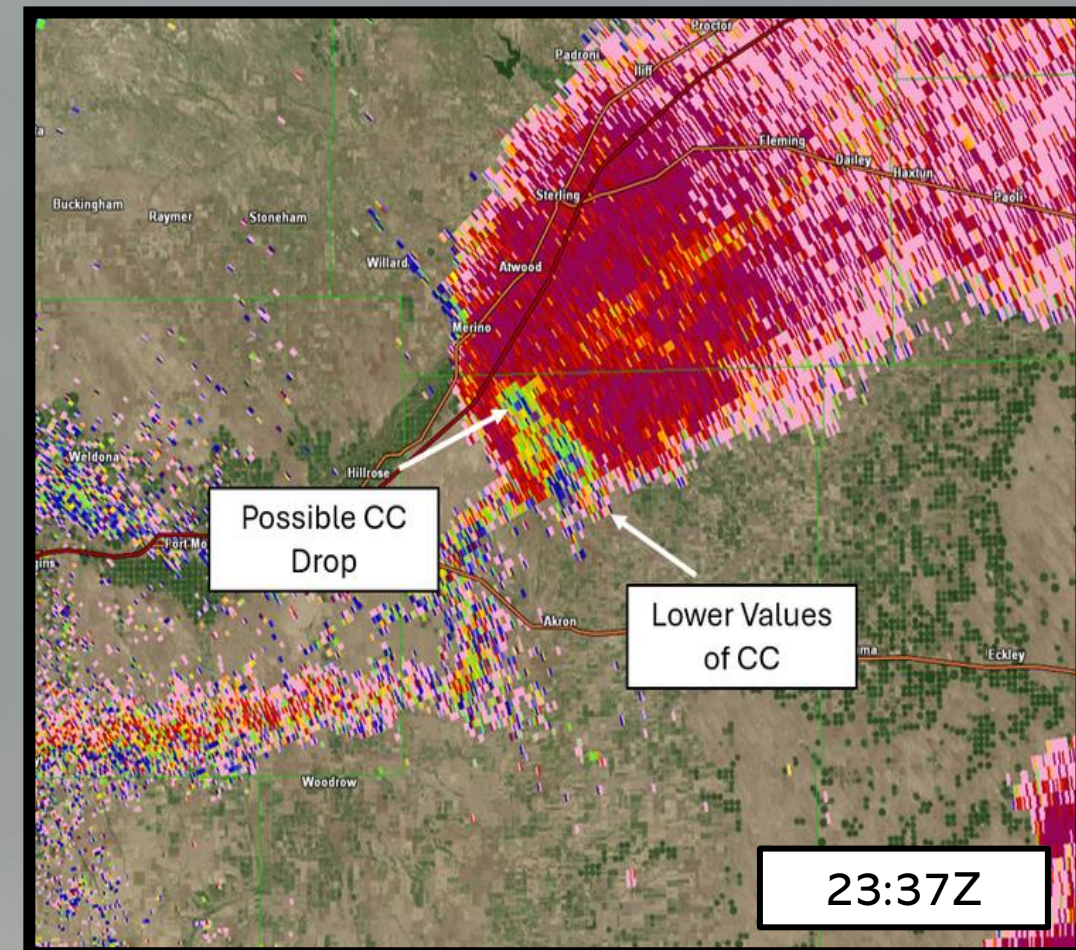
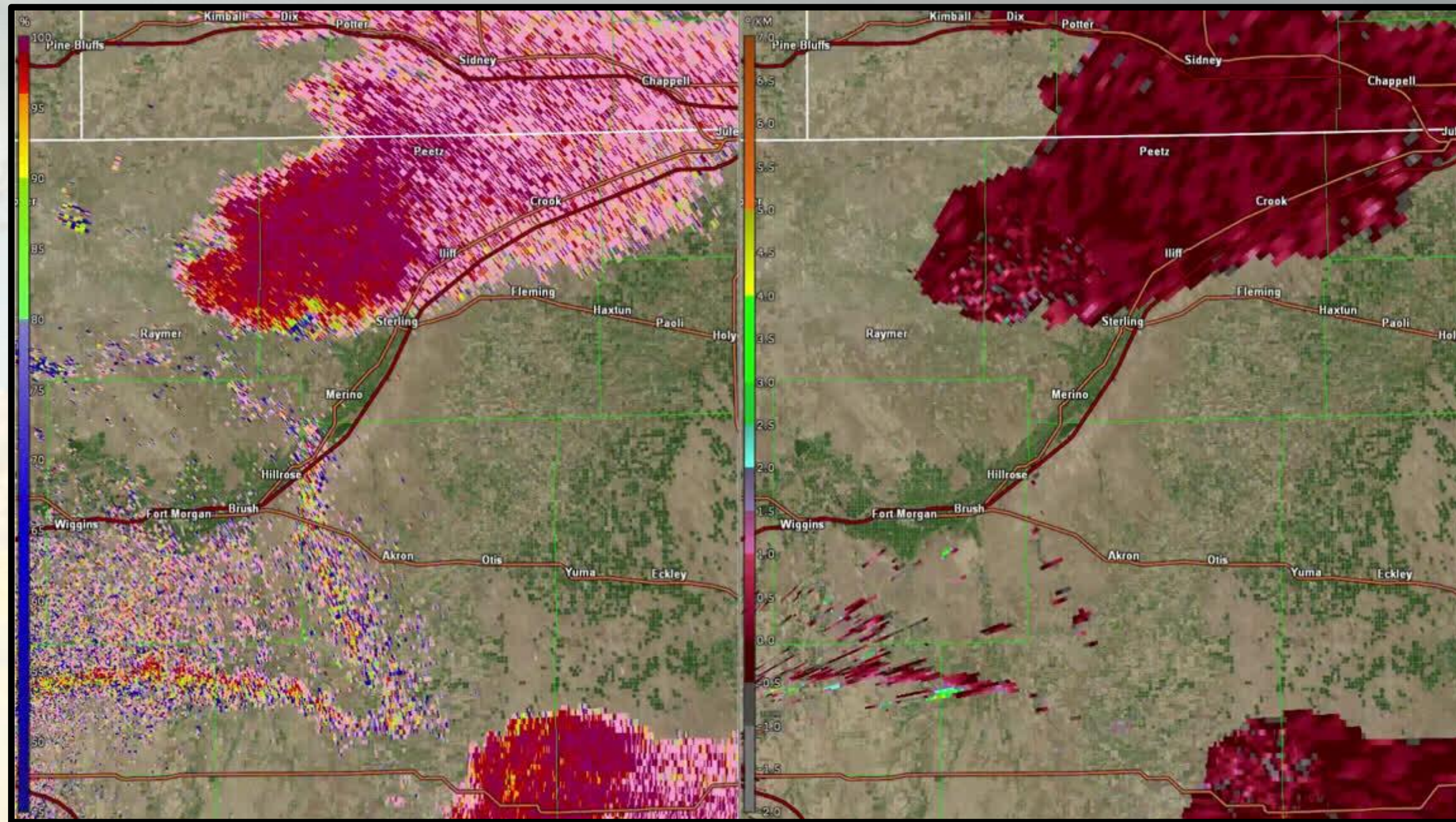


RADAR – SPECTRUM WIDTH



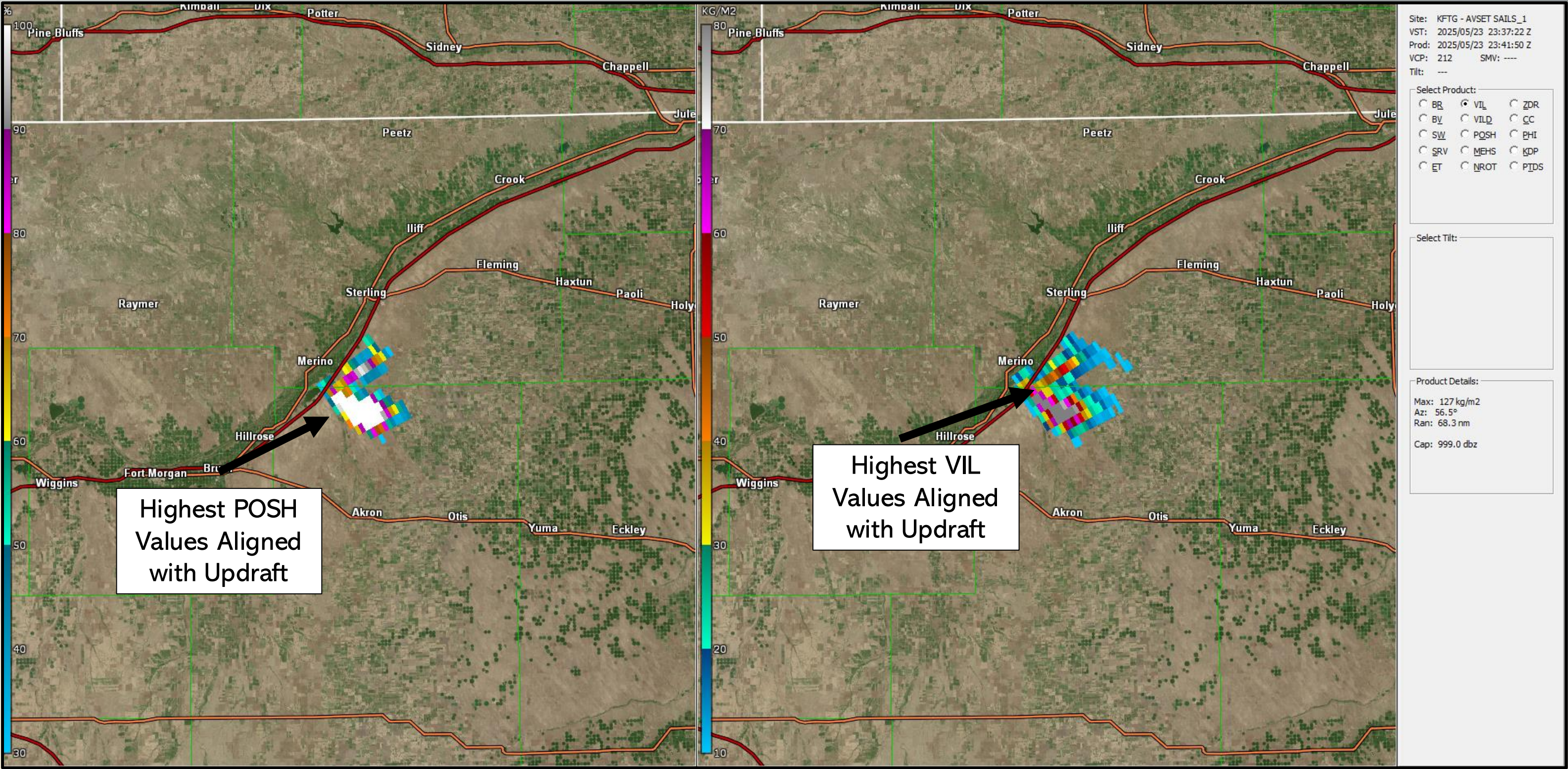
- High turbulent flow near the updraft/mesocyclone
- OFB storm is attached to also sees increased values of turbulent flow

RADAR – CORRELATION COEFF. & KDP



- CC drop at 2337z suggests possible debris/hail
- Enhanced KDP column indicative of growing hydrometeors, high liquid water content near updraft

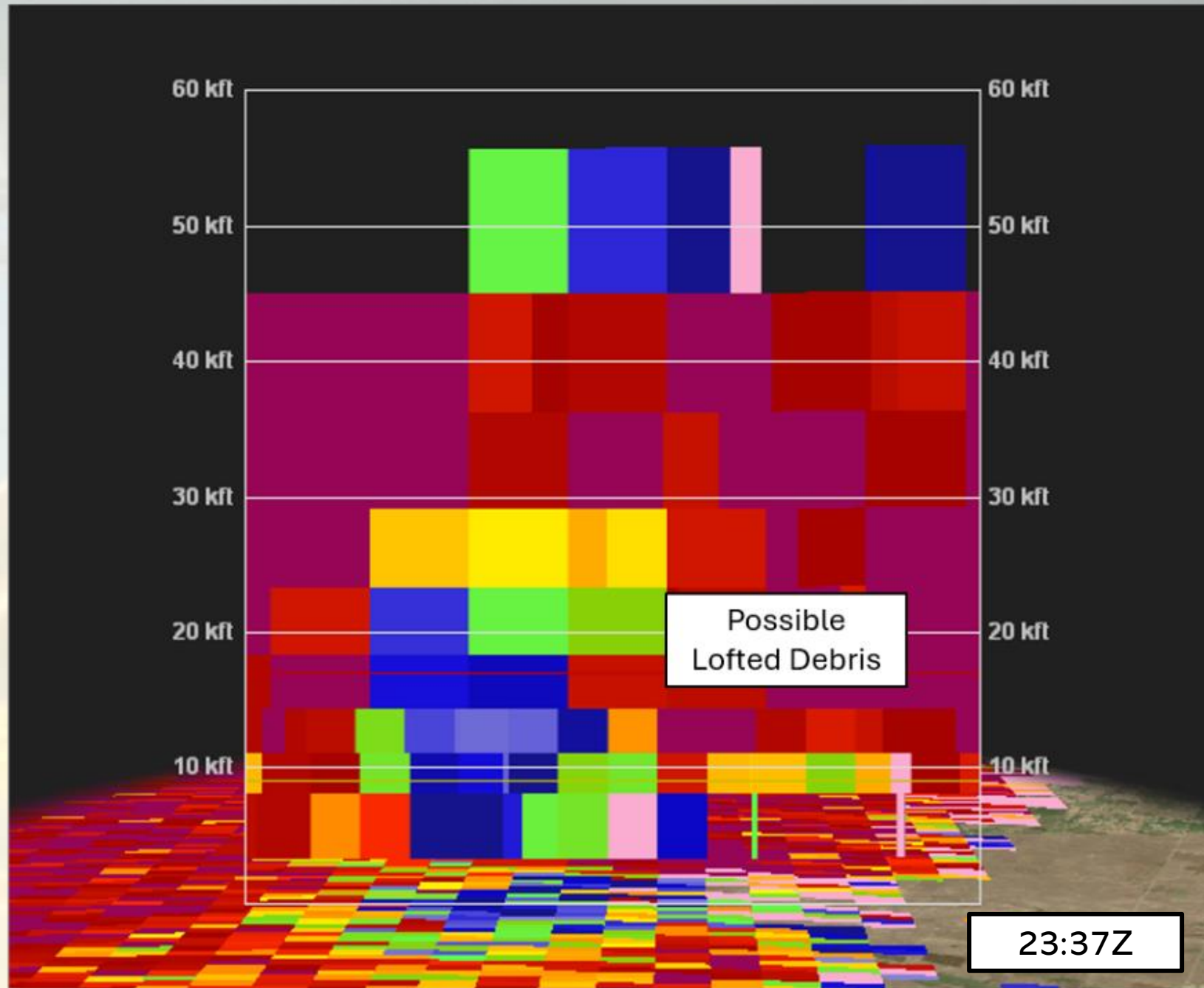
RADAR – HAIL PARAMETERS



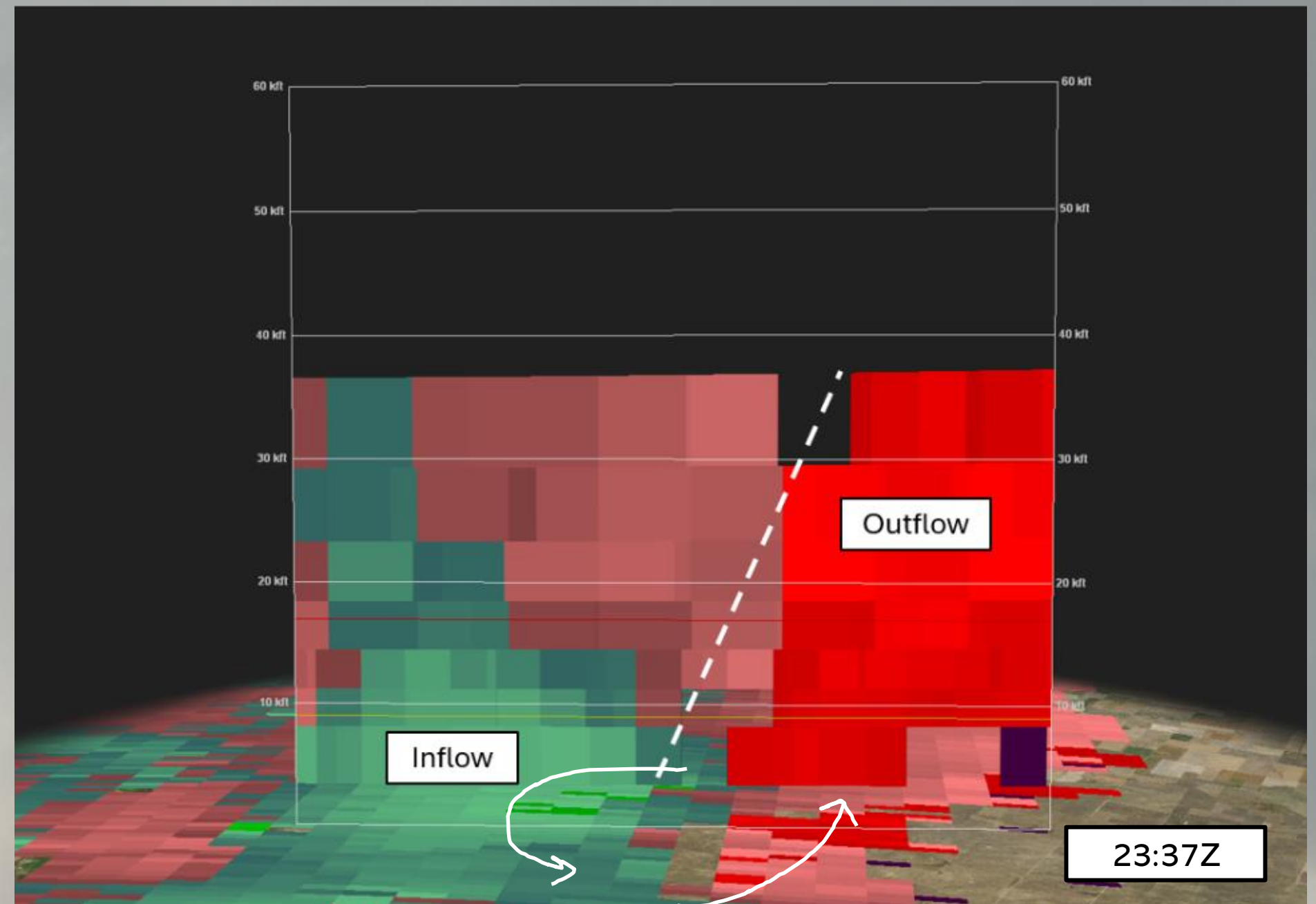
- High POSH values indicative of a strong probability of severe hail

- High VIL values indicative of a deep updraft (consistent with hail growth)

RADAR – CROSS SECTIONS



- Lower CC values indicative of lofted debris ~20kft



- Tilted updraft consistent with tornadic development at 2337z

RADAR – CROSS SECTIONS

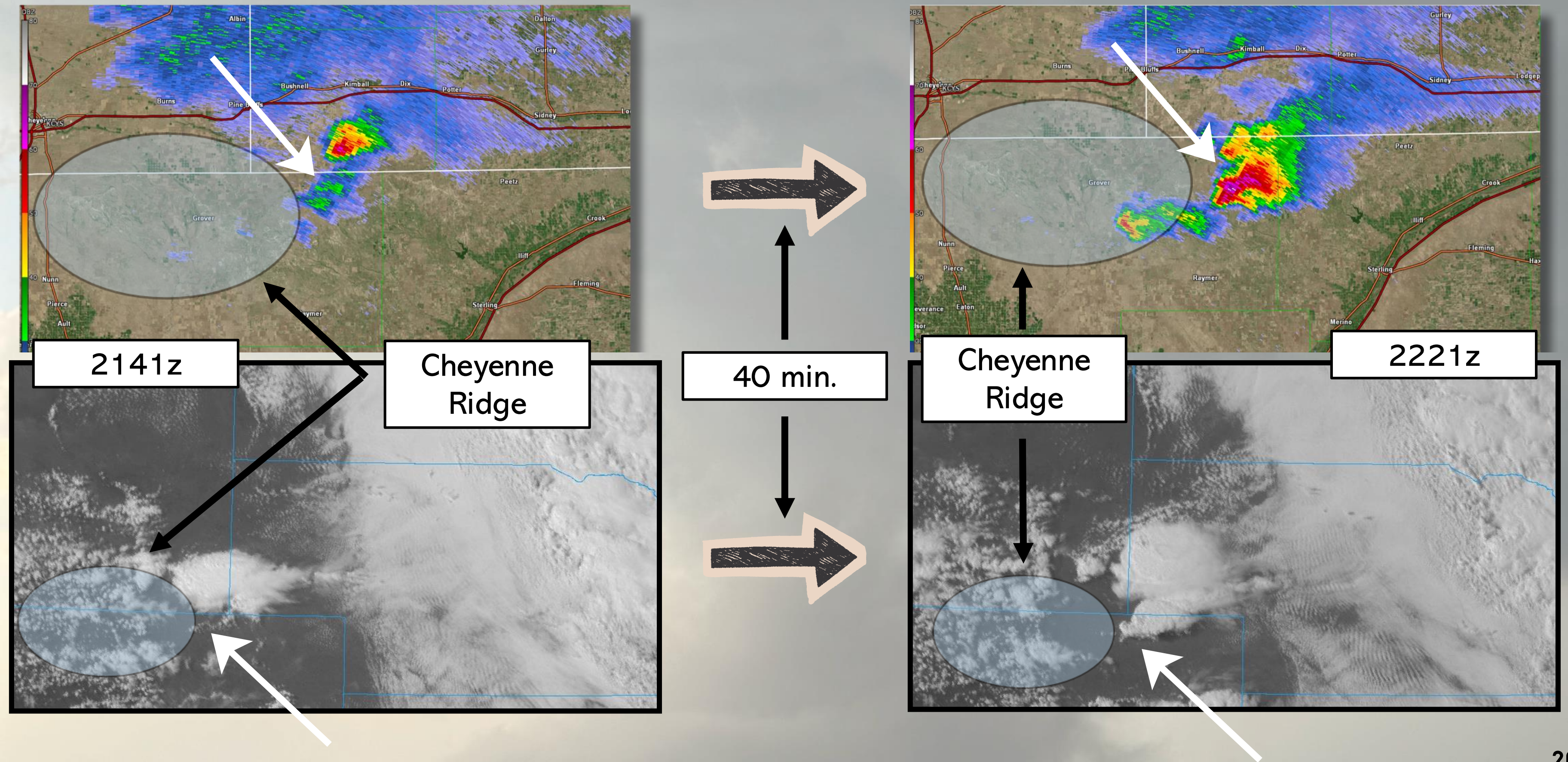


- Tilted updraft influenced by values of shear (bottom displaced from top)
- Defined, textbook hook echo



- Cloud tops exceeding 40kft indicative of strong & sustained updraft
- High ref. values indicative of large hydrometers aloft

RADAR/SATELLITE – TERRAIN INFLUENCE



CONCLUSIONS

1. **Synoptic setup:** weak but sufficient UL support, with Akron in the right-entrance region of the jet streak and small height falls aiding ascent.
2. **Mesoscale setup:** favorable factors, with CAPE (SFC), lower LCLs, and enhanced shear/SRH directly over NE CO
3. Terrain influences from the Cheyenne Ridge enhanced LL flow and convergence, which focused storm initiation/intensification as the supercell moved off the ridge.
4. **Satellite signatures** (OTs, V-notch, cloud top temps, LLWV moisture gradients) showed a powerful sustained updraft and strong storm development leading up to the tornado.
5. **Radar data** (SRV, spectrum width, CC drops, VIL/POSH, 3D cross-sections) showed a strengthening LL circulation, turbulent flow, hail, and a textbook supercell structure at the peak intensity.

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THANK YOU!



Questions?